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Foraging Paralysis as a Collapse Mechanism

A Bifurcation Analysis of Honey Bee Colony Dynamics
under Predation by *Vespa velutina*

Bachelor's Thesis in Mathematics

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Abstract

The invasive yellow-legged hornet *Vespa velutina* has spread across Europe since 2004 and preys on honey bee colonies at the hive entrance. Field observations show that its presence also induces foraging paralysis. Bees reduce or stop foraging, far out of proportion to the bees actually taken. This behavioural pathway has been documented empirically and explored through agent-based simulation, but, to my knowledge, not in a model that yields explicit thresholds.

This thesis places the hornet in a three-compartment model of hive bees, foragers and food stores, as an external parameter acting through two channels: the direct loss of foragers to predation and the reduction of foraging activity. Both channels are calibrated against the same field study, so the thresholds come out in hornets rather than in an abstract stress parameter.

The equilibria follow in closed form, and every positive equilibrium of the delay-free model is proved to be locally stable. An exact energy budget of the colony yields a Lyapunov function that proves collapse under sustained pressure. Three hornet loads organise the results. Above about six hornets a well-stocked colony stops gaining stores (numerically), above about eight the positive equilibrium no longer exists and above about ten every model colony converges to extinction. With the mortality of a paralysed forager held at its baseline value, all three lie inside the range of loads observed in the field.

Predation, the mortality channel, barely matters: in the calibrated model it moves every threshold by about a hundredth of a hornet. Foraging paralysis alone is sufficient to drive a model colony to collapse.

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Notation

Table 1: Symbols, units and reference values. Sources of the parameter values: Appendix A.

Symbol	Meaning	Unit	Value
H	hive bees	bees	state
F	foragers	bees	state
f	food stores	g	state
B	brood (not a state, see Sec. 4.4)	bees	LS/φ
N	$H + F$, adult population	bees	–
L	maximum laying rate	eggs d ⁻¹	2000
$\alpha_{\min}$	minimum recruitment rate	d ⁻¹	0.25
$\alpha_{\max}$	food-dependent recruitment	d ⁻¹	0.25
σ	social inhibition	d ⁻¹	0.75
v	half-saturation, brood care	bees	5000
b	half-saturation, food	g	500
φ	pupation rate	d ⁻¹	1/9
τ	capped (pupal) stage	d	12
c, c_0	food per forager, calibrated value c_0 , Rem. 6.1	g d ⁻¹	0.10
γ_A	adult food consumption	g bee ⁻¹ d ⁻¹	0.007
γ_B	brood food consumption	g bee ⁻¹ d ⁻¹	0.018
m_0	baseline forager mortality, 1/6.5 rounded	d ⁻¹	0.154
ν	foraging trips per day	d ⁻¹	3
ℓ	load per trip, c/ν	g	0.0333
V	hornet load	hornets	parameter
$p(V)$	relative foraging activity	–	$e^{-\beta V}$
β	decay of foraging activity	hornet ⁻¹	0.109
$\text{HF}(p)$	homing failure per return flight	–	$h_0 e^{-\zeta p}$
h_0, ζ	prefactor and exponent of HF	–	0.0020, 6.09
$\mu(V)$	additional forager mortality	d ⁻¹	$\nu p \text{HF}(p)$
m	$m_0 + \mu(V)$, total mortality	d ⁻¹	–
θ	flight-related share of m_0 , Sec. 6.3	–	0 (bracket [0, 1])
κ	γ_B/φ , honey per bee reared	g	0.1620
P	H/F allowed by the food balance, Eq. (5.2)	–	–
J	$1/P = F/H$ at equilibrium	–	–
Q	H/F selected by recruitment, full combs, Eq. (5.5)	–	2.8334
P_c	H/F at the collision, $mv/(Ls^*)$, Prop. 5.6	–	1.3667
s^*	$s(f_{\text{crit}})$, brood survival at the critical store level	–	0.2817
p_X	balance threshold with X hive bees charged, Eq. (5.30)	–	–
$s(f)$	$f^2/(f^2 + b^2)$	–	–
$\alpha(f)$	$\alpha_{\min} + \alpha_{\max} b^2/(b^2 + f^2)$	d ⁻¹	–
u	F/N , forager fraction	–	–
W_a	Lyapunov function with weight a , Lemma 5.14	g	$f + a(H + F)$
p_{acc}	accumulation boundary, upper end of the band	–	0.5178
p^*	existence boundary, lower end of the band	–	0.4151
a_*	$\kappa + \gamma_A v/L$, largest admissible common weight	g	0.1795
p_E	collapse bound: collapse proved for $p < p_E$	–	0.3464
p_κ	break-even bound (weight κ)	–	0.3195
a_c	$\kappa + \gamma_A v/(Ls^*)$, weight for low stores, Prop. 5.33	g	0.2241
m^*	existence boundary in m at $p = 1$	d ⁻¹	0.4624
f_{crit}	critical store level: stability change on the food axis	g	313.1
V_{acc}	hornet load – $\ln p/\beta$ at p_{acc}	hornets	6.04
V^*	hornet load at p^*	hornets	8.07
V_E	hornet load at p_E	hornets	9.73

Chapter 1

Introduction

This chapter describes the attribution problem that motivates the thesis, introduces *V. velutina* as a stressor with two pathways, states the research questions and summarises the results.

1.1 Colony losses and the attribution problem

Managed honey bee colonies die over winter at rates that beekeepers and national surveys have tracked for two decades. What those surveys cannot do is say why. A colony that fails in February was exposed to mites, to pesticide residues, to a poor forage year and to a queen of uncertain quality. It failed once.

The modelling literature has taken this seriously rather than working around it. Bryden et al. [6] show that chronic sublethal stress, which impairs individual bees rather than killing them outright, can still bring a colony down. Booton et al. [5] give the mechanism its shape. A stressor that raises forager mortality moves a colony across a fold. The collapse is then sudden, while the stress that caused it was gradual. Dennis and Kemp [10] put the same behaviour in terms of an Allee effect: below a critical colony size the growth rate per bee becomes negative, so a small colony declines further. The common conclusion is that colony failure is a threshold phenomenon. A threshold crossed by several stressors at once cannot be apportioned among them afterwards.

This thesis does not attempt an apportionment. It takes a single stressor whose two pathways were measured separately in one field campaign. It then asks which of the two does the work. That question has an answer, where the general attribution question does not.

1.2 *V. velutina* as a behaviourally mediated stressor

The yellow-legged hornet *V. velutina* was recorded in southwest France in 2004 and is the first vespid predator to reach Europe from Asia [29]. It hunts by hovering in front of a hive entrance and taking foragers in flight (Section 2.2).

The obvious way to model a predator is as a mortality term and that is what the literature has done. Requier et al. [37] quantified something else, a response Monceau et al. [30] had reported from two hives. Watching colonies under natural hornet pressure, they recorded two distinct effects. Foraging traffic collapsed as the hornet load rose, by roughly an order of magnitude between zero and twenty hornets. And bees that did fly failed to return at a rate that, in the fitted relation, never exceeds one flight in five hundred.

The second effect is predation. The first is not. A colony under hornet pressure stops foraging and it stops far out of proportion to the bees actually lost. Whether that behavioural response can kill a colony on its own is the first question of this thesis.

The question is worth asking because the two pathways point in opposite directions for management. If the hornet kills by eating bees, the response is to kill hornets. If it kills by frightening a colony out of its own foraging season, the load at which that happens may be far below the load at which the losses look alarming. The damage then shows up months later, as a colony that went into winter light. The response is then to act at lower loads, before losses become visible, and to judge a control measure by whether it keeps the colony foraging, not only by how many hornets it removes.

1.3 Research questions

- Q1** Can foraging paralysis alone, without any increase in mortality, drive a colony to collapse at hornet loads inside the observed range?
- Q2** Is there a threshold and can it be stated explicitly and interpretably?
- Q3** Which of the two channels can reach the existence boundary, the load at which the colony's positive equilibrium disappears, at field-realistic hornet loads?

In a model with a food balance the qualitative part of Q1 is structural: a colony that collects nothing starves. What Q1 asks is at which load and whether that load lies inside the zero to twenty hornets that Requier et al. [37] observed. Q2 asks about form. Q3 needs one yardstick for both channels. Like the model of Khoury et al. [22], the model of this thesis has no comb capacity. An undisturbed model colony therefore has no *positive equilibrium*, that is, no equilibrium with positive populations: in the integrations its stores grow without bound. A sufficiently large hard capacity would move neither the existence boundary nor the collapse bound (Section 5.1.2). A positive equilibrium appears only under pressure and disappears again at the existence boundary. A channel that cannot reach it can neither remove the positive equilibrium nor carry the colony to the collapse bound beyond it, so both channels are measured against it.

1.4 Overview of the results

Foraging paralysis alone is sufficient, in the model and under sustained pressure. At the reference parameter values of Table 1 and with forager mortality held at its undisturbed value, three loads organise the answer. Above about six hornets a well-stocked colony stops gaining stores (numerically). Above eight its only positive equilibrium disappears. Above 9.73 every model colony is proved to converge to extinction, also when an empty comb forces it to live on its daily intake (Section 5.3.3). Near the bound, that convergence outlasts a hornet season (Section 5.3.3, Table B.7). The first two are the sharp boundaries of the existence band and the third a sufficient bound. All three are explicit and come in a provable order (Section 5.5). For the mortality of a paralysed forager they are the pessimistic end of a bracket: if a forager kept in the nest were spared her flight-related mortality, they would move to about eleven, seventeen and twenty-one hornets, the last just beyond the observed loads (Section 6.3). For forage they are optimistic, since the intake rate is a plentiful-forage value and poorer autumn forage lowers all three (Table 4.1, assumption (g)). Predation through mortality moves every threshold by about a hundredth of a hornet (Section 6.2). Figure 1.1 collects these loads.

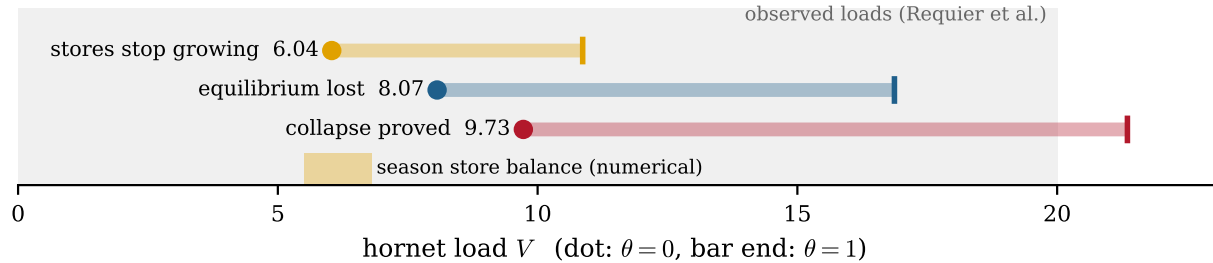


Figure 1.1: The three hornet loads that organise the results: the accumulation boundary at 6.04 hornets (5.6), below which a well-stocked colony gains stores (numerically), the existence boundary at 8.07 (Proposition 5.6) and the collapse bound at 9.73 (Theorem 5.17). Dots: forager mortality fixed at m_0 . Bar ends: all of it scales with flight activity (Section 6.3). Grey: loads observed by Requier et al. [37]. Short bar: season store-balance loads, numerical (Section 6.4).

1.5 Outline

Chapter 2 gives the biology the model needs. Chapter 3 presents the models of Khoury et al. [21, 22]. Chapter 4 places the hornet in the second of them and calibrates both channels against Requier et al. [37]. Chapter 5 contains the analysis, from the equilibrium cascade to the collapse bound, Chapter 6 compares the two channels and Chapter 7 discusses and concludes. The appendices carry the provenance of every number, the numerical protocols, the reproduction of the published model with two corrections and the longer proofs. Numerical results are controls, not proofs, and are marked as such.

Chapter 2

Biological background

This chapter collects the biology on which the model rests: the division of labour between hive bees and foragers, the hunting behaviour of *V. velutina*, the consequences of reduced foraging and the defences of the two honey bee species that *V. velutina* meets.

2.1 Division of labour and social inhibition

A worker honey bee changes job with age. She spends her first weeks inside the nest, cleaning cells, feeding larvae and processing nectar. Then she becomes a forager for the last part of her life [38]. The transition is not on a fixed clock. It is regulated by the colony.

Two features of that regulation matter for the model.

The first is social inhibition. Foragers returning to the nest slow the maturation of younger bees, through direct contact and through ethyl oleate, which they produce and probably pass on with food [25]. The more foragers a colony has, the more slowly it makes new ones. If foragers are lost the inhibition weakens and hive bees mature faster, so the colony repairs its own age structure without any central control [17, 38]. Conversely, foragers confined to the hive delay the maturation of the younger bees [17].

The second is that the transition runs both ways. A colony that has lost too many hive bees can send foragers back to nest duties and this reversion has been observed when foragers become a large part of the colony [17, 38]. Both directions are needed, because a model with only forward maturation cannot recover from a shock to the hive bees.

Food enters this regulation as well. Under food shortage a colony accelerates the maturation of its young bees and puts them out to forage sooner [46, 51]. A colony that is short of stores therefore pushes harder, which is the response that makes the food dynamics interesting rather than passive.

2.2 *V. velutina*: predation behaviour and seasonality

V. velutina hunts honey bees at the hive entrance, seizing foragers in flight from the traffic that hovers there [29]. Beekeepers call the behaviour hawking. It is not a raid on the nest. The hornet takes bees in the air, in the small volume through which every forager has to pass.

The pressure is seasonal. Monceau et al. [28] monitored hornet pressure at two apiaries through a season and found predation lasting more than five months, from summer into December, with the peak in the autumn. That is when the hornet colony rears its sexual brood, which needs animal protein [29, 37].

The timing is the part of the biology that makes a small number of hornets matter. A honey bee colony in October is not foraging for the day. It is laying in the stores it will live on until spring. Pressure applied in those weeks does not have to kill anything to be expensive.

Hawking also explains why the effect is not confined to the bees that are caught. The hornets occupy the entrance, which is the one place every forager must cross, so the whole traffic sees them.

2.3 Consequences of reduced foraging

That reduced activity can harm a colony is not obvious and it has been measured. Rojas-Nossa et al. [41] protected colonies with electric harps and compared them with unprotected ones. The protected colonies kept higher foraging activity, brought in more pollen, reared more brood, produced heavier workers and survived the winter better. The harps remove the hornets and with them both pathways of Section 4.2, so the study shows that hornet pressure is costly and that foraging activity moves with colony fate. Its authors read the result as support for foraging paralysis as the main reason for the weakening of colonies under attack. The design does not separate paralysis from predation and that separation is what the model is for. This thesis models the first link of that chain, foraging activity. The stores f stand for the rest. It does not model the winter itself.

2.4 Defences of *A. cerana* and *A. mellifera*

A. cerana, the Asian honey bee, has shared a range with *V. velutina* for a long time. Its workers engulf an attacking hornet in a ball and heat her to death [32]. *A. mellifera* balls with fewer bees, at a lower core temperature and without the wing shimmering of *A. cerana* guards [19, 49]. In a shared apiary the hornets' hawking success against it was about three times higher, 31 % against 11 % [49]. What *A. mellifera* mainly does is stop flying [37]. That looks like a defence and is not one: a colony that stops foraging in October gives up the income it needs for the winter.

Chapter 3

The Khoury models

This chapter presents the two models this thesis builds on and explains why the second of them is the natural starting point for a stressor that acts on foraging rather than on survival. The source of every parameter value is in Appendix A. Reproducing both models (Appendix C) turned up two errors in the appendix of Khoury et al. [22] and an extinction threshold that is sufficient but does not mark the loss of the equilibrium, all checkable against figures in the same paper.

3.1 The 2011 model: hive bees and foragers

Khoury et al. [21] reduce a colony to two classes of adult workers. Hive bees H tend brood and maintain the nest, foragers F work outside and $N = H + F$. Hive-bee mortality is neglected, foragers die at rate m and brood is not a state:

$$\frac{dH}{dt} = L \underbrace{\frac{N}{w + N}}_{\text{eclosion}} - \underbrace{H R(H, F)}_{\text{recruitment}}, \quad \frac{dF}{dt} = H R(H, F) - mF, \quad R(H, F) = \alpha - \sigma \frac{F}{H + F} \quad (3.1)$$

Eclosion saturates at the laying rate L , with half the maximum at w adults. Recruitment carries the social inhibition of Section 2.1: α is the rate at which hive bees become foragers when no foragers are present. The term linear in the forager fraction F/N lets foragers hold that maturation back.

One property of R matters later: R *changes sign*. Since F/N ranges over $[0, 1]$, R takes values in $[\alpha - \sigma, \alpha]$ and $R < 0$ exactly when $F/N > \alpha/\sigma = \frac{1}{3}$. Above a forager fraction of one third the net flow reverses and foragers return to hive duties.

3.1.1 Parameter values

The laying rate is $L = 2000$ eggs per day [9]. A worker can forage from the age of four days [13] and Khoury et al. [21] take the reciprocal, $\alpha = 0.25 \text{ d}^{-1}$. Their choice $\sigma = 3\alpha = 0.75 \text{ d}^{-1}$ makes reversion begin above a forager fraction of one third, a modelling assumption. Reversion itself is documented in colonies depleted of young bees [38].

The forager death rate is the reciprocal of the flightspan, the number of days a bee is seen foraging. Khoury et al. [21, Tab. 1] list four colonies with flightspans of 7.5, 6.5, 6.7 and 8.8 days [43]. This thesis uses the shortest, $m_0 = 0.154 \text{ d}^{-1}$ ($1/6.5 \text{ d}$ rounded, with every threshold below computed with 0.154). That is the least favourable case for a colony. Across all four colonies the behavioural thresholds move by 1.5 to 2.1 hornets and the comparison of the two channels is unaffected (Table A.2). A honey bee lives five or six weeks, but only the last week of that is spent foraging: the parameter that governs the whole model is the length of a bee's final job.

3.1.2 Two features carried forward

In the phase portraits of Khoury et al. [21, Fig. 3] trajectories collapse quickly onto a ray $F = JH$ of fixed ratio and then move slowly along it: the age structure settles fast, the population size slowly. The sign change of R also carries into the 2013 model, where the food-dependent term

raises the recruitment rate up to $\alpha_{\min} + \alpha_{\max} = 0.5$ and R changes sign at every food level because $\sigma = 0.75 > \alpha_{\min} + \alpha_{\max}$.

3.2 The 2013 model: adding food

3.2.1 The equations

Khoury et al. [22] add two compartments (Figure 3.1). The uncapped brood B holds eggs and larvae. The stores f hold food, measured in grams, with pollen and nectar not separated. Capped brood is not a state. The time a pupa spends in its cell enters as a delay instead:

$$\begin{cases} \frac{dB}{dt} = L S(H, f) - \varphi B, & \text{laying and survival, minus pupation} \\ \frac{dH}{dt} = \varphi B(t - \tau) - H R(H, F, f), & \text{eclosion, minus recruitment} \\ \frac{dF}{dt} = H R(H, F, f) - mF, & \text{recruitment, minus death} \\ \frac{df}{dt} = cF - \gamma_A (H + F) - \gamma_B B, & \text{intake, minus consumption} \end{cases} \quad (\text{K})$$

with

$$S(H, f) = \underbrace{\frac{f^2}{f^2 + b^2}}_{=: s(f)} \cdot \frac{H}{H + v}, \quad R(H, F, f) = \underbrace{\alpha_{\min} + \alpha_{\max} \frac{b^2}{b^2 + f^2}}_{=: \alpha(f)} - \sigma \frac{F}{F + H} \quad (3.2)$$

Brood survival. S is a product of two factors. The food factor $s(f)$ is sigmoid, not Michaelis–Menten. The reason is biological. When stores run low, brood starve and workers eat young larvae themselves, recycling the protein into the older larvae that still have a chance [45]. A colony under food stress does not lose brood gradually. It culls. The hive-bee factor $H/(H + v)$ models nursing capacity: too few hive bees cannot feed all the larvae or hold the brood nest at the temperature the pupae need [18].

Pupation, recruitment and food. Larvae pupate at the constant rate φ and pupae spend τ days capped with negligible mortality, so the bees emerging today entered pupation τ days ago. Recruitment gains the food-driven acceleration $\alpha_{\max} b^2 / (b^2 + f^2)$ of Section 2.1. It uses the same half-saturation constant b as $s(f)$, which encodes an assumption worth naming: the shortage the larvae feel and the shortage that pushes bees outdoors are the same shortage. Intake is cF . Consumption is charged to brood at γ_B and to both adult classes at one common rate γ_A , an average over hive bees and foragers that this thesis keeps.

Two further changes against 2011 go unremarked in the 2013 paper. Brood rearing now depends on hive bees alone, through $H/(H + v)$ in brood survival, rather than on all adults through eclosion. The half-saturation constant falls from $w = 27\,000$ adults to $v = 5\,000$ hive bees. In 2011 a forager still counted towards the colony’s ability to rear brood, but in 2013 she does not.

3.2.2 Parameter values

L , $\alpha_{\min}$ and σ are inherited from 2011. Four of the new values record a modelling decision rather than a measurement. The recruitment amplitude $\alpha_{\max} = 0.25 \text{ d}^{-1}$ makes recruitment double when food is absent, $\alpha(0) = 0.5$ against $\alpha(\infty) = 0.25$, a factor chosen by Khoury et al. [22]: Schulz et al. [46] show the acceleration, not its size. The food half-saturation $b = 500 \text{ g}$ makes the effects of low stores vanish above a kilogram. The nurse half-saturation $v = 5000$

bees is the hive-bee population at which roughly half the eggs laid fail to reach pupation [22], a value at the critical colony strength rather than comfortably above it (Appendix A.1). The brood consumption $\gamma_B = 0.018$ g per day is the 0.163 g of honey that Harbo [15] measured for rearing one worker to the pupal stage, averaged over the nine uncapped days. The intake $c = 0.1$ g per forager per day is the June maximum of Russell et al. [44], from data of Harbo [14], a plentiful-forage value. The remaining values are $\gamma_A = 0.007$ g per day [15], $\varphi = 1/9$ d⁻¹ and $\tau = 12$ d.

3.2.3 Choice of the 2013 model

Foraging paralysis reduces what a forager brings home without killing her. In (3.1) there is no food compartment and no collection rate: the quantity paralysis acts on does not occur. Attaching p by hand to L , α or m would prescribe the consequence of lost intake instead of deriving it and that consequence is the question of this thesis. In the 2013 model the lost honey enters the food balance and its consequences follow from the model's own structure.

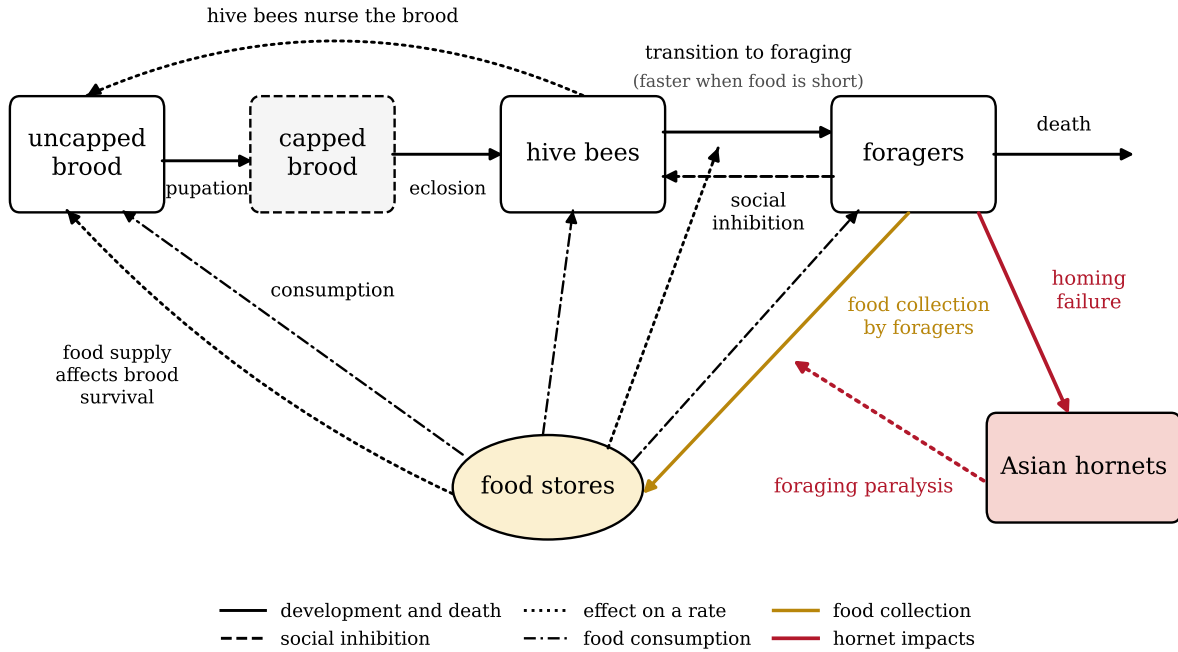


Figure 3.1: Colony dynamics of (K), after Fig. 1 of Khoury et al. [22]. Capped brood (dashed) is not a state but enters as the delay τ . The two hornet impacts are the two channels of Chapter 4: homing failure ($m_0 \mapsto m_0 + \mu(V)$) and foraging paralysis ($c \mapsto cp(V)$).

3.3 Biological interpretation

A healthy forager flies for about 6.5 days. The 2011 model says a colony fails once that falls below $1/0.355 = 2.8$ days [21]. Any stressor that shortens the foraging life by more than half is therefore enough, whatever the stressor is. That single number is what the later literature builds on and it is why forager mortality became the axis everyone reaches for.

The 2013 model adds the reason that axis is not the only one. A forager brings home 0.1 g of honey a day and eats 0.007 g, so she earns about fourteen times her own keep. Rearing her replacement costs 0.163 g once and since she must be replaced every 6.5 days that is $\kappa m_0 = 0.025$ g per day. With the upkeep of the 2.8 hive bees that stand behind each forager in a well-fed colony, she spends about half of what she collects. The whole model turns on this per-day balance. In good conditions, stores stop accumulating once activity falls by about half.

Chapter 4

Relocating the stressor

Picture the entrance board of a colony in October. Hornets hover in front of it, waiting for returning foragers. Some bees are caught. Most are not. What the colony does is stop flying.

That difference is the whole of this chapter. A forager who stays inside is alive. She brings nothing home, but she still eats, she still counts in the adult population and she can go back to nursing. Mortality and intake are different quantities and a stressor that reduces intake without killing anyone has no representation in a mortality parameter. This chapter states the model that keeps them apart, places the two hornet effects in it and calibrates both against one field study.

4.1 The model

I take Khoury et al. [22] unchanged except at two symbols, in its delay-free three-dimensional form:

$$\begin{cases} \frac{dH}{dt} = L S(H, f) - H R(H, F, f), \\ \frac{dF}{dt} = H R(H, F, f) - [m_0 + \underbrace{\mu(V)}_{\text{channel 1}}] F, \\ \frac{df}{dt} = c \underbrace{p(V)}_{\text{channel 2}} F - \gamma_A (H + F) - \frac{\gamma_B L}{\varphi} S(H, f), \end{cases} \quad (\text{M})$$

with S and R as in (3.2). The substitutions are $c \mapsto c p(V)$ and $m_0 \mapsto m_0 + \mu(V)$. I write $m := m_0 + \mu(V)$ throughout. The brood term $\gamma_B L S / \varphi$ is $\gamma_B B$ with $B = L S / \varphi$ (Section 4.4).

Placement of the two channels. Channel 1 is homing failure. A hornet removes a returning forager, which is a death, so it raises the forager death rate and enters the F -equation. Channel 2 is paralysis. A colony flying less brings in the fraction $p(V)$ of what it would otherwise collect, so it multiplies the intake term and enters the f -equation. Laying rate, brood survival, recruitment and all consumption rates stay as published, so every difference between the equilibria of this analysis and Khoury's is attributable to two symbols.

V is a parameter, not a state. The number of hornets in front of a colony is set by nearby hornet nests and by the landscape, not by the colony. So (M) contains no hornet dynamics: it is a one-parameter family of vector fields indexed by V .

4.2 Two impacts, one of them not a death

Requier et al. [37] document both impacts of the hornet and it matters that they are two.

- **Homing failure (HF):** a hornet catches a returning forager and carries her off. The bee is lost. This is predation and it belongs in the death rate.

- **Foraging paralysis (FP):** with hornets at the entrance, the colony cuts its flight traffic. Requier et al. [37] measure it as flight activity relative to its maximum and report that in the worst events the bees “completely cease their flight traffic at the hive entrance”.

The two impacts are not independent. Requier et al. [37] find that homing failure *rises* as flight activity falls: a hornet picks off bees more easily from a thin stream than from a dense one. This inverts what a defence should do. The colony’s response to a predator makes the predator’s job easier. Requier et al. [37] call the paralysis “an unadapted behavioural response to the invasive predator” (compare Section 2.4).

Section 7.1 compares this placement of the two impacts with the earlier stressor models.

4.2.1 Limits of the substitution

Requier et al. [37] feed paralysis into the BEEHAVE colony model [3] by reducing the maximal foraging distance with the simulated load, to zero at the highest loads (Section 7.2). A paralysed forager then stays in the nest and is no longer exposed to the risks of foraging, so at high loads the adult population *rises* and the colony dies over winter because that larger population has eaten the honey. In (M) a paralysed forager keeps dying at the baseline rate m_0 . Only the hornet-caused part of her mortality scales with p (Section 4.3.4). The model therefore does not reproduce the rise in population and collapse runs through the energy budget alone. Which reading of m_0 is closer to the field is not known. This thesis treats the choice as a bracket: the model as written is its pessimistic end and Section 6.3 gives the other end and what it costs in hornets. Two unquantified features point the same way. A paralysed forager keeps eating at the rate γ_A , part of which may be flight fuel. Falling stores still push hive bees through $\alpha(f)$ into a class that cannot fly but dies at m_0 . The paralysed forager also keeps her published roles: she counts in the social inhibition $\sigma F/(H + F)$, as confined foragers do [17]. She does not nurse, since brood care depends on H alone. Both are modelling decisions and the second works against the colony.

4.3 Calibration of the two hornet functions

$$p(V) = e^{-\beta V}, \quad \beta = 0.109 \text{ hornet}^{-1}, \quad \mu(V) = \nu p(V) \text{ HF}(p(V)), \quad \text{HF}(p) = 0.0020 e^{-6.09 p} \quad (4.1)$$

Both functions are calibrated against Requier et al. [37] alone and Table 4.1 collects the assumptions made along the way.

4.3.1 The hornet load V and its source

Requier et al. [37] define the hornet load by how they measured it: an observer records, for fifteen minutes, the maximum number of hornets hovering in front of one entrance at the same time and then counts returning bees for two minutes, which gives the flight activity (Appendix A.2). Throughout this thesis, V is that quantity.

The field literature offers at least three incompatible metrics for hornet pressure: Rojas-Nossa et al. [41] count hornets per colony per ten minutes and Diéguez-Antón et al. [11] count the hornets present in hourly photographs of the entrance. A curve fitted to one metric cannot be evaluated at a load measured in another, so a threshold in hornets means nothing without its metric. The other field studies therefore appear in this thesis as qualitative support, never as data.

4.3.2 The paralysis function $p(V)$

Requier et al. [37, Fig. 2a] plot predicted flight activity as a percentage of its maximum against hornet load, from a Poisson generalized linear mixed model (Appendix A.2), presumably as the

Table 4.1: Assumptions of the calibration, what each reads into the data and which way a failure would move the thresholds. Assumptions (a) to (f) do not affect the thresholds in p . They act only on the conversion to hornets or on the size of the mortality channel. Assumption (g) scales every threshold in p by c_0/c , with $c_0 = 0.1 \text{ g d}^{-1}$ the calibrated intake and c the intake in the hornet season (Remark 6.1).

assumption	why it is needed	if it fails
(a) The decline in traffic is a response to the hornets.	Loads above five hornets occur only between 28 August and 6 November, when activity also falls for seasonal reasons (Table A.5).	β is too large and the thresholds in hornets err low.
(b) The decline is fewer trips per forager, not fewer foragers.	The survey counts whole-colony traffic, while forager depletion is produced by (M) itself and would otherwise be counted twice.	Same direction as (a).
(c) Paralysis reduces the number of trips, not the load per trip.	The survey counted flights, not loads.	Intake falls faster than traffic and the thresholds in hornets err high.
(d) A constant V is sustained all day and all season.	V is a fifteen-minute peak count, applied to the whole foraging day.	If daily loads lie below the peak, the thresholds err low.
(e) The load acts on the risk per flight only through traffic.	The video data come from one colony with at most four hornets in the frame.	μ may be underestimated, but step 2 of Section 6.2 does not use this assumption.
(f) The activity axis of Fig. 2b has the scale of Fig. 2a. One colony in one season is representative.	Fig. 2b comes from the video colony and Fig. 2a from 131 colonies. Captures outside the frame are missed.	A scale factor k changes (4.5) by $1/k$. The estimate is biased downwards, so Section 6.2 states every margin as a factor.
(g) The intake $c = 0.1 \text{ g}$ per day holds in the hornet season.	June maximum of Russell et al. [44], from Harbo [14]. October gives $0.45 c$.	Every threshold in hornets falls by $\ln(c_0/c)/\beta$, about 7 in October.

prediction at random effects zero. The paper does not say. A Poisson model uses a log link, so a load entering the linear predictor linearly gives a response exponential in V :

$$p(V) = e^{-\beta V}, \quad \text{equivalently} \quad -\ln p(V) = \beta V \quad (4.2)$$

A least-squares fit through the origin on the digitised values (Table B.1, Appendix B.1) gives 0.1098, and the value used in this thesis is $\beta = 0.109$ per hornet. The pointwise estimates spread over $[0.107, 0.120]$. That spread measures the digitisation, not the statistical uncertainty of the fitted curve, which is wider: the lower confidence bound of Requier et al. [37, Fig. 2a] reaches zero activity at 12.6 hornets, although the fitted curve has no zero (Remark A.2). Section 6.3 therefore also carries an envelope of $\pm 20\%$.

4.3.3 The homing-failure function $\mu(V)$

The second channel comes from the video part of the same study, recorded at one colony near La Rochelle in October 2015 (Appendix A.2). For each 15-minute slot the homing failure is the number of bees caught divided by the number of incoming bee trajectories. It depends significantly and negatively on flight activity.

The axis of Requier et al. [37, Fig. 2b] is flight activity, not hornet load. That is why HF is written as a function of activity and composed with $p(V)$ afterwards. It is the formal expression of the coupling described in Section 4.2. Requier et al. [37] fit a binomial model with a logit link. For probabilities below 0.2% the logistic curve and the exponential agree to within 0.2% of their value. This thesis uses $h_0 = 0.0020$ and $\zeta = 6.09$, so that $\text{HF}(p) = h_0 e^{-\zeta p}$. The exponent lies

inside the pointwise interval $[6.01, 6.14]$ of the digitisation (Table B.2, Table A.5). The limits are $\text{HF}(0) = 0.20\%$ and, extrapolated beyond the digitised range $p \leq 0.5$, $\text{HF}(1) = 4.5 \cdot 10^{-6}$: the risk per return flight spans between two and three orders of magnitude between an undisturbed colony and a fully paralysed one.

4.3.4 Foraging trips and exposure

HF is a probability per return flight, a pure number, while μ must be a rate per day. The conversion needs the number of trips per day and whether the hornets change it.

The hornets change the trip rate, which fixes the form of μ . Requier's flight activity counts returning bees. The model reads its decline as fewer flights with an unchanged load per flight. Writing ν for the undisturbed trip rate and ℓ for the load per trip,

$$c = \nu \ell, \quad (4.3)$$

and the substitution $c \mapsto cp(V)$ says that ν falls to $\nu p(V)$ while ℓ stays. The decomposition is not invented here: Perry et al. [34] write the food equation of the same model family with a load per trip times a number of trips per day. A forager that flies less often is exposed less often, because HF is charged per flight. Hence

$$\underbrace{\mu(V)}_{\text{d}^{-1}} = \underbrace{\nu p(V)}_{\text{trips d}^{-1}} \cdot \underbrace{\text{HF}(p(V))}_{\text{per trip}} \quad (4.4)$$

Leaving the factor $p(V)$ out would make the two substitutions disagree. The food equation would describe a colony that flies less and the mortality equation a colony that flies as much as ever.

The value. From their RFID flight data Perry et al. [34] report that “the mean number of trips per day for all bees (including those that did not complete any foraging trips) was approximately three” and obtain their load per trip of 0.033 g by exactly the decomposition (4.3), 0.1 g over three trips. Reading their average over all bees as the convention of Khoury's c , every bee in the forager class on every day, is an interpretation. I adopt $\nu = 3 \text{ d}^{-1}$.

Direct measurements bracket the value: per bee and foraging day, 2.9 trips in five pooled studies [39], 4.6 in Minahan and Brunet [27], about four in Klein et al. [23] (17 to 21 trips over 4.2 to 4.9 days) and about seven implied by Visscher and Dukas [52]. Since μ is linear in ν , Table A.4 gives the consequence of each.

The mortality channel is bounded. With $\text{HF}(p) = h_0 e^{-\zeta p}$, (4.4) reads $\mu = \nu h_0 p e^{-\zeta p}$. The function $p \mapsto p e^{-\zeta p}$ vanishes at $p = 0$, attains its maximum at $p = 1/\zeta$ and decreases afterwards, so

$$\mu(V) \leq \frac{\nu h_0}{\zeta e} = 3.62 \cdot 10^{-4} \text{ d}^{-1} = 0.235\% \text{ of } m_0 \quad \text{for every } V \geq 0, \quad (4.5)$$

with equality at $p = 1/\zeta = 0.164$, that is at $V = 16.6$ hornets. Table A.4 gives the bound for other trip rates and Table A.3 evaluates the channel at the thresholds.

Remark 4.1 (The mortality channel is numerically invisible in the thresholds). Chapter 5 states its thresholds at $m = m_0$. Recomputing them with $m = m_0 + \mu(V)$ raises each by at most 0.0005 in p and lowers it by about a hundredth of a hornet (Section 6.2). The collapse statements hold exactly: $p_E(m)$ increases with m and $\mu \geq 0$, so $p(V) < p_E(m_0)$ implies $p(V) < p_E(m_0 + \mu(V))$.

4.4 The brood compartment and the delay

The published model (K) is four-dimensional and carries a delay, whereas (M) is neither. The delay appears only in front of B , so eliminating B removes the fourth dimension and the delay together. The elimination is a modelling choice and not a separation of time scales: with $1/\varphi = 9$ d against $\tau = 12$ d no quasi-steady-state argument is available.

Remark 4.2 (Equilibria are τ -independent). A constant solution of a delay equation satisfies $x(t - \tau) = x(t)$, so the delay drops out of the equilibrium conditions. Applied here it gives $\varphi B = LS(H, f)$, so the eclosion term $\varphi B(t - \tau)$ equals LS and the brood consumption $\gamma_B B$ equals $\gamma_B LS/\varphi$, for every $\tau \geq 0$. Substituting both into (K) reproduces (M). The equilibria of the four-dimensional delay model and of (M) coincide.

The remark says nothing about stability and that is where the reduction earns its keep: (M) gives a 3×3 Jacobian (Section 5.2) instead of a transcendental characteristic equation. The argument for (M) is economy and not accuracy, as is usual in this literature [e.g. 5, 21]. The global collapse result does not need it (Theorem 5.20). Numerically, the two place the finite-horizon survival threshold within 0.6 % of each other at the longest horizon tested, against 2.7 % at 1000 d, and the gap closes as the horizon grows (Table B.3). The linearised delay model is stable at six points across the band, with relaxation up to 1.7 times slower (Table 5.1, Appendix B.2), but its basins are not tested. Of the finite-time results, the one that carries the field interpretation is repeated in the delay model (Table 6.3).

4.5 Well-posedness, positivity and boundedness

The right-hand side of (M) extends to a locally Lipschitz field of at most linear growth on the set of non-negative states $\Omega = \{(H, F, f) : H, F, f \geq 0\}$, so solutions exist, are unique and do not blow up in finite time. The faces $\{H = 0\}$ and $\{F = 0\}$ cannot be crossed outwards (Appendix D.9). The face $\{f = 0\}$ can. There $\dot{f} = cpF - \gamma_A(H + F)$, which is negative whenever intake falls below adult consumption. The model, like Khoury's, has no mechanism that rations consumption when the comb is empty, so a solution that reaches $\{f = 0\}$ with $\dot{f} < 0$ leaves the regime the equations describe, at any load (Remark 5.21), although the field itself extends to $f < 0$. Statements about solutions of (M) hold while they remain in Ω , the *model's domain*. Proposition 5.19 closes the gap. The population is bounded at every load (Corollary 5.24), the stores for $p < p_E$ (Section 5.3). In between that is open and for $p > p_{acc}$ they grow without limit in the integrations.

4.6 Biological interpretation

Five hornets cut the flight traffic by about two fifths and ten cut it to a third. The response is steepest below five hornets, where most of Requier's observations lie. A handful of insects in front of the entrance changes what a colony of thirty thousand does with its day. Against that, even in the thinnest traffic the fitted rate never exceeds one return flight in five hundred ending in a hornet. A colony loses very few bees to being eaten. It loses its reserves by not flying.

Chapter 5

Equilibria and stability

Chapter 4 handed over a three-dimensional ODE system with two hornet functions calibrated against field data. This chapter asks three questions of that system. Where are the positive equilibria? Are they stable? And what happens to a colony that does not start at one?

The equilibria fall out of a *cascade* rather than a root-find. Local stability follows from a Jacobian whose structure the equilibrium relations simplify drastically. And the global statement starts from an *identity*, the colony's energy budget, from which the Lyapunov function is built with a single estimate rather than guessed. Figure 5.1 shows which result is used where.

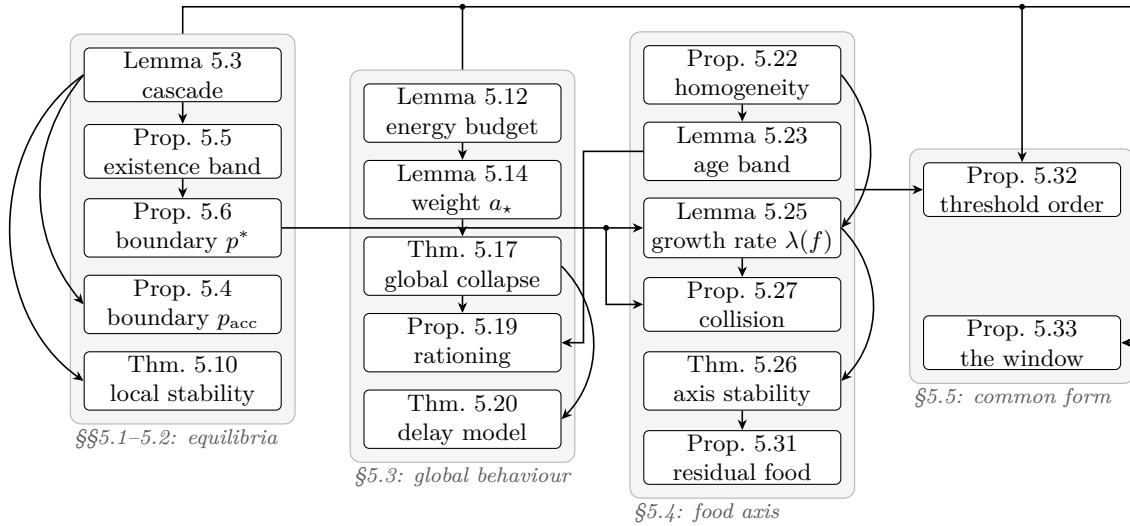


Figure 5.1: Main lines of proof in this chapter. An arrow means “is used in the proof of”. An arrow leaving a whole panel stands for several of its results.

The chapter yields four thresholds in the activity p , all at $m = m_0$: the *accumulation boundary* p_{acc} , the *existence boundary* p^* , the *collapse bound* p_E and the *break-even bound* p_K . Table 5.2 in Section 5.5 collects and orders them.

Definition 5.1 (Collapse and persistence). A solution *collapses* if $H(t) \rightarrow 0$ and $F(t) \rightarrow 0$ as $t \rightarrow \infty$. It *persists* if $\limsup_{t \rightarrow \infty} (H(t) + F(t)) > 0$.

5.1 The equilibrium cascade

Setting the right-hand side of (M) to zero gives three algebraic equations in three unknowns. Solved as a system they are unpleasant: S and R are both nonlinear in two variables each. Solved in the right order they are three one-line problems and each step uses one structural feature of the model: the food balance fixes the age ratio, food-dependent recruitment fixes the food level and saturating eclosion fixes the size. Each is needed: without saturation no size is fixed and with $\alpha_{max} = 0$ step (ii) fixes the ratio twice.

Lemma 5.2 (Adult balance). Consider any model of the form $\dot{H} = E - HR$, $\dot{F} = HR - mF$, in which hive-bee mortality is neglected, where E denotes the eclosion rate and R the recruitment

rate. Then at every equilibrium $E = mF$ holds. The lemma also covers the delay differential equation (K), in which $E = \varphi B(t - \tau)$ and τ is the length of the capped stage. There it reads $LS(H, f) = mF$, for every $\tau \geq 0$.

Proof. Adding the two equations gives $\dot{H} + \dot{F} = E - mF$. The recruitment term HR cancels, because every bee recruited leaves H and enters F , whatever the sign of R . At an equilibrium the left-hand side vanishes. For (K) the eclosion rate is $E = \varphi B(t - \tau)$. A constant solution satisfies $B(t - \tau) = B$ and $\dot{B} = 0$ gives $\varphi B = LS(H, f)$. $\square$

Applied to (M), eclosion equals forager loss at every equilibrium,

$$E = LS = mF \quad (5.1)$$

Every term in the food equation is therefore proportional to F . Intake is cpF , adult consumption is $\gamma_A(H + F)$ and brood consumption is $\gamma_B B = \kappa LS = \kappa mF$. So F cancels and the food balance fixes a *ratio* rather than a level.

Lemma 5.3 (The cascade). *Let (H^*, F^*, f^*) be an equilibrium of (M) with $H^*, F^*, f^* > 0$. Then, with $\kappa = \gamma_B/\varphi$ and $J := F^*/H^*$,*

$$(i) \text{ food balance } \quad \frac{1}{J} = P := \frac{cp}{\gamma_A} - 1 - \frac{\gamma_B m}{\gamma_A \varphi}, \quad (5.2)$$

$$(ii) \text{ recruitment } \quad \alpha_{\max} \frac{b^2}{b^2 + (f^*)^2} = mJ + \frac{\sigma J}{1 + J} - \alpha_{\min}, \quad (5.3)$$

$$(iii) \text{ eclosion } \quad H^* = \frac{Ls(f^*)}{mJ} - v, \quad F^* = JH^* \quad (5.4)$$

The three equations are solved in this order: (5.2) gives J from the parameters, (5.3) gives f^* from J and (5.4) gives H^* from f^* and J . Each step is a one-line problem given the previous ones. The three unknowns are not independent of each other.

Proof. (i) By (5.1), $\gamma_B LS/\varphi = \kappa mF$. Substituting into $\dot{f} = 0$,

$$cpF = \gamma_A(H + F) + \kappa mF \iff \gamma_A H = F(cp - \gamma_A - \kappa m),$$

and dividing by $\gamma_A F$ gives $H/F = cp/\gamma_A - 1 - \kappa m/\gamma_A$. Since $\kappa/\gamma_A = \gamma_B/(\gamma_A \varphi)$, this is (5.2).

(ii) Setting $\dot{F} = 0$ gives $HR = mF$, hence $R = mJ$. Writing R out with $F/(H + F) = J/(1 + J)$,

$$\alpha_{\min} + \alpha_{\max} \frac{b^2}{b^2 + f^2} - \frac{\sigma J}{1 + J} = mJ,$$

which rearranges to (5.3). The left-hand side of (5.3) is strictly decreasing in f , from $\alpha_{\max}$ at $f = 0$ to 0 as $f \rightarrow \infty$, so f^* is unique when it exists.

(iii) $LS = mF$ with $S = s(f)H/(H + v)$ and $F = JH$ gives $Ls(f)H/(H + v) = mJH$, so $Ls(f) = mJ(H + v)$, which is (5.4). $\square$

5.1.1 P is set by the budget, Q by recruitment

Equation (5.2) holds only where $\dot{f} = 0$. At a positive equilibrium P is the realised ratio H^*/F^* . At the reference parameters with $p = 1$ and $m = m_0$ it gives $P = 9.7217$, so approximately ten hive bees per forager. The colony does not run there. In the food-abundant regime the realised ratio is $Q = 2.8334$, that is 29.1 % of P .

The two numbers answer different questions. P is the largest hive-bee-to-forager ratio the energy budget can *sustain* once brood production has settled to replacement, $LS = mF$: any

more hive bees and their upkeep exceeds what the foragers bring in. Q is the ratio that recruitment *selects* when food is not scarce. With full combs, $\alpha(f) \rightarrow \alpha_{\min}$ and step (ii) reduces to

$$\alpha_{\min} - \frac{\sigma J}{1+J} = mJ \iff mJ^2 + (m + \sigma - \alpha_{\min})J - \alpha_{\min} = 0, \quad Q = \frac{1}{J}, \quad (5.5)$$

whose positive root at $m = m_0$ is $J_Q = 0.352932$, hence $Q = 2.8334$.

The two meet exactly at the boundary of the food-limited regime.

Proposition 5.4 (The food-limitation boundary). *The expression $mJ + \sigma J/(1+J) - \alpha_{\min}$ on the right-hand side of (5.3) vanishes if and only if $J = J_Q$, that is $P = Q$.*

Proof. Multiplying by $1+J > 0$ gives $mJ^2 + (m + \sigma - \alpha_{\min})J - \alpha_{\min} = 0$, which is (5.5). Its constant term is negative, so it has exactly one positive root, J_Q . $\square$

Since $\alpha(f) \rightarrow \alpha_{\min}$ as $f \rightarrow \infty$, this expression is the quantity that must vanish for f^* to escape to infinity. This gives p_{acc} in closed form:

$$P(p_{\text{acc}}, m_0) = Q(m_0) \implies p_{\text{acc}} = \frac{\gamma_A}{c} \left(Q + 1 + \frac{\gamma_B m_0}{\gamma_A \varphi} \right) = 0.517818, \quad (5.6)$$

that is $V_{\text{acc}} = 6.04$ hornets. In the m -direction the same boundary is located at $m = 0.4010$, consistent with Khoury et al. [22, Fig. 2b/2c].

5.1.2 The existence band

Lemma 5.3 yields an equilibrium only when each step is solvable. Collecting the conditions gives a statement sharper than a threshold.

Proposition 5.5 (Existence). *Fix $m > 0$. A positive equilibrium of (M) exists if and only if all three of the following hold:*

1. $P > 0$, that is $cp > \gamma_A + \kappa m$: the foragers pay for themselves.
2. $0 < mJ + \frac{\sigma J}{1+J} - \alpha_{\min} < \alpha_{\max}$: step (ii) has a solution $f^* \in (0, \infty)$.
3. $Ls(f^*) > mJv$: step (iii) gives $H^* > 0$.

At $m = m_0$ the set of admissible p is the open interval

$$(p^*, p_{\text{acc}}) = (0.415150, 0.517818), \quad \text{that is} \quad V \in (6.04, 8.07) \text{ hornets}, \quad (5.7)$$

where the lower boundary is condition 3 becoming an equality and the upper boundary is condition 2 failing at its lower bound.

The equivalence is Lemma 5.3 read backwards. That the admissible set is an interval follows from monotonicity in p at fixed m . As p increases, P increases and J decreases. The expression $mJ + \sigma J/(1+J) - \alpha_{\min}$ is increasing in J , so it decreases, f^* increases and $s(f^*)$ increases, while mJv decreases. Condition 3 is therefore monotone in p and holds above a single value p^* . The lower bound in condition 2 holds exactly while $J > J_Q$, that is below p_{acc} . The upper bound holds above a value that lies below p^* : as this expression approaches $\alpha_{\max}$, $f^* \rightarrow 0$ and $s(f^*) \rightarrow 0$, so condition 3 has already failed there. Condition 1 holds on the whole band. The upper boundary is (5.6). The lower boundary follows from the cascade alone.

Proposition 5.6 (The existence boundary in closed form). *Fix $m > 0$, put $\omega := L/v$ and let $s^* = s^*(m)$ be the positive root of*

$$(\omega^2 + \omega\alpha_{\max})s^2 + [\omega(\sigma + m - \alpha_{\min} - \alpha_{\max}) + m\alpha_{\max}]s - m(\alpha_{\min} + \alpha_{\max}) = 0 \quad (5.8)$$

Put

$$f_{\text{crit}} := b\sqrt{\frac{s^*}{1-s^*}}, \quad J_c(m) := \frac{Ls^*}{mv}, \quad P_c(m) := \frac{1}{J_c(m)} = \frac{mv}{Ls^*} \quad (5.9)$$

Then at the lower boundary of the existence band $f^ = f_{\text{crit}}$, $J = J_c$ and*

$$p^*(m) = \frac{\gamma_A(1 + P_c(m)) + \kappa m}{c} = \frac{1}{c} \left[\gamma_A \left(1 + \frac{mv}{Ls^*} \right) + \kappa m \right] \quad (5.10)$$

Proof. Steps (i) and (ii) of Lemma 5.3 define J and f^* as continuous functions of p up to and including the lower boundary, where conditions 1 and 2 hold strictly, as shown above. The boundary is where step (iii) gives $H^* = 0$, that is $Ls(f^*) = mJv$. With $A := \omega s(f^*)$ this reads $A = mJ$, while step (ii) reads $\alpha(f^*) = mJ + \sigma J/(1 + J)$. Eliminating $J = A/m$ gives

$$A^2 + A(\sigma + m - \alpha) - \alpha m = 0 \quad (5.11)$$

With $A = \omega s$ and $\alpha = \alpha_{\min} + \alpha_{\max} - \alpha_{\max}s$ this is (5.8). Its constant term is negative and its leading coefficient positive, so it has exactly one positive root and $s(f^*) \in (0, 1)$ is that root. Hence $f^* = f_{\text{crit}}$ and $J = A/m = J_c$. Step (i) at the boundary, $1/J_c = cp^*/\gamma_A - 1 - \kappa m/\gamma_A$, gives (5.10). Dividing (5.11) by m , with $A = mJ$, shows that J_c is the positive root of $mJ^2 + (m + \sigma - \alpha(f_{\text{crit}}))J - \alpha(f_{\text{crit}}) = 0$. $\square$

At $m = m_0$ this gives $s^* = 0.281699$, $f_{\text{crit}} = 313.12$ g, $J_c = 0.7317$, $P_c = 1.3667$ and $p^* = 0.415150$, the value of (5.7). At $m = m^*$ it gives $s^* = 0.447317$ and $f_{\text{crit}} = 449.82$ g. The boundary moves right as mortality grows: $p^*(m)$ is strictly increasing (Appendix D.6). Equation (5.11) returns in Section 5.4 as the condition for a neutrally stable point of the food axis (Proposition 5.27). That is why the lower boundary is a collision and not a fold.

The undisturbed colony. The band is narrow and it does not contain the undisturbed colony. At every $p > p_{\text{acc}}$, including $p = 1$, the population settles in the integrations at the age ratio Q while the stores grow without bound, as in Khoury et al. [22, Fig. 2b]. That is a property of a model with no comb capacity and no season, not a claim about hives (a hard capacity above f_{crit} would not move f_{crit} , p^* or p_E). The loss of the positive equilibrium of Proposition 5.5 at V^* is therefore a statement about the long-run geometry of the model and the field meaning of V_{acc} comes from the season (Section 6.4).

Remark 5.7 (The two boundaries of the band). The two boundaries fail differently and only the lower one is a bifurcation. At p_{acc} conditions 1 and 3 hold and the middle expression of condition 2 falls to zero. That expression is $\alpha_{\max}\psi$ with $\psi = b^2/(b^2 + f^2)$, so its vanishing means $f^* \rightarrow \infty$: the equilibrium does not disappear, it escapes to infinite stores and the colony is doing well. At p^* conditions 1 and 2 hold with room to spare, so J and f^* have limits with $J > 0$ and $0 < f^* < \infty$. Condition 3 fails because H^* reaches zero. The equilibrium collides with the food axis $\{H = F = 0\}$ (Proposition 5.27).

This thesis calls the lower boundary a *collision with the food axis* (Proposition 5.27). The term transcritical is avoided on purpose. There the trivial branch changes stability at the parameter value where the other branch crosses it. Here the stability change along the axis lies at f_{crit} for every p (Section 5.4) and what p^* changes is the direction of the slow drift along the axis at that point (Remark 5.28).

Corollary 5.8 (Existence boundary in m). *At $p = 1$, condition 3 of Proposition 5.5 fails at $m^* = 0.4624055 \text{ d}^{-1}$, that is at 3.00 times m_0 , an increase of 0.3084 over the baseline.*

This is the mortality at which the positive equilibrium *ceases to exist* at full foraging activity, not the mortality at which a colony provably dies, which lies further out (Theorem 5.17). It follows from the closed form of p^* by setting $p^*(m) = 1$ (Section 5.5).

5.2 Local stability by linearisation

5.2.1 The Jacobian

Write $s = s(f)$, $\psi = b^2/(b^2 + f^2) = 1 - s$, $g = H/(H + v)$, $u = F/(H + F)$ and $s'(f) = 2fb^2/(f^2 + b^2)^2$. These abbreviations are functions of the state (H, F, f) , as are S and R , and are evaluated at the point at which the Jacobian is taken. The prime denotes the derivative with respect to f . Differentiating (M) and using $\alpha'(f) = -\alpha_{\max} s'(f)$ gives

$$\mathcal{J} = \begin{pmatrix} Ls \frac{v}{(H+v)^2} - R - \frac{\sigma HF}{(H+F)^2} & \frac{\sigma H^2}{(H+F)^2} & s'(Lg + H\alpha_{\max}) \\ R + \frac{\sigma HF}{(H+F)^2} & -\frac{\sigma H^2}{(H+F)^2} - m & -H\alpha_{\max} s' \\ -\gamma_A - \kappa Ls \frac{v}{(H+v)^2} & cp - \gamma_A & -\kappa Lg s' \end{pmatrix} \quad (5.12)$$

The characteristic polynomial of $\mathcal{J}$ is $\det(\lambda I - \mathcal{J}) = \lambda^3 + a_1\lambda^2 + a_2\lambda + a_3$, with $a_1 = -\text{tr } \mathcal{J}$, a_2 the sum of the three principal 2×2 minors of $\mathcal{J}$ and $a_3 = -\det \mathcal{J}$. Written with the entries of (5.12), these coefficients are of considerable size. The equilibrium relations cut that down drastically.

Lemma 5.9 (Reduced trace). *At a positive equilibrium of (M),*

$$a_1 = -\text{tr } \mathcal{J} = \underbrace{mJ \frac{H}{H+v}}_{\text{brood-care saturation}} + \underbrace{\frac{\sigma}{1+J}}_{\text{social inhibition}} + \underbrace{m}_{\text{forager turnover}} + \underbrace{\frac{2\kappa m J H \psi}{f}}_{\text{brood cost}} \quad (5.13)$$

In particular $a_1 > m > 0$ at every positive equilibrium.

The proof is algebra on the entries of the Jacobian and is given in Appendix D.1. Every term is manifestly positive and each one names a stabilising mechanism: no compartment of this model carries a destabilising self-feedback.

5.2.2 Every positive equilibrium is stable

The same equilibrium relations that reduce the trace reduce the matrix entries. At a positive equilibrium, $LS = mF$ gives $Lsg = mJH$, step (ii) of Lemma 5.3 gives $R = mJ$ and step (i) gives $cp - \gamma_A = \kappa m + \gamma_A/J$. With $s'(f) = 2s\psi/f$, the Jacobian (5.12) becomes

$$\mathcal{J} = \begin{pmatrix} -mJg - \frac{\sigma J}{(1+J)^2} & \frac{\sigma}{(1+J)^2} & Hs' \left(\frac{mJ}{s} + \alpha_{\max} \right) \\ mJ + \frac{\sigma J}{(1+J)^2} & -m - \frac{\sigma}{(1+J)^2} & -\alpha_{\max} Hs' \\ -\gamma_A - \kappa mJ(1-g) & \kappa m + \frac{\gamma_A}{J} & -\frac{\kappa mJHs'}{s} \end{pmatrix} \quad (5.14)$$

All three diagonal entries are negative and the off-diagonal pairs $(\mathcal{J}_{13}, \mathcal{J}_{31})$ and $(\mathcal{J}_{23}, \mathcal{J}_{32})$ each have entries of opposite sign. The proof of Theorem 5.10 uses this sign pattern to bound a_2 from below.

By the Routh–Hurwitz criterion [31, Appendix B], all roots of $\lambda^3 + a_1\lambda^2 + a_2\lambda + a_3$ have negative real part if and only if $a_1 > 0$, $a_3 > 0$ and $a_1a_2 > a_3$. An equilibrium at which these

conditions hold is locally asymptotically stable, since all eigenvalues of its Jacobian then have negative real part.

Theorem 5.10 (Local stability). *At every positive equilibrium of (M), for all positive parameter values,*

$$a_3 = -\det \mathcal{J} = m \gamma_A \alpha_{\max} \frac{H}{H+v} H s'(f) > 0 \quad \text{and} \quad a_1 a_2 > a_3 \quad (5.15)$$

Hence every positive equilibrium is locally asymptotically stable.

Proof. The determinant. Adding the second row of (5.14) to the first and then κ times the new first row to the third, does not change the determinant. The new first row is the gradient of $LS - mF = \dot{H} + \dot{F}$, namely $(mJ(1-g), -m, mJHs'/s)$. The new third row is the gradient of $\dot{f} + \kappa(\dot{H} + \dot{F}) = cpF - \gamma_A(H+F) - \kappa mF$, namely $(-\gamma_A, \gamma_A/J, 0)$, since $cp - \gamma_A - \kappa m = \gamma_A/J$. Expanding along the third column, the cofactor of the first entry is

$$\left(mJ + \frac{\sigma J}{(1+J)^2}\right) \frac{\gamma_A}{J} - \left(m + \frac{\sigma}{(1+J)^2}\right) \gamma_A = 0,$$

and the cofactor of the second entry is $-[mJ(1-g)\gamma_A/J - m\gamma_A] = m\gamma_A g$. Hence $\det \mathcal{J} = -\alpha_{\max} H s' \cdot m\gamma_A g$, which is the first claim.

The second coefficient. By definition, a_2 is the sum of the three principal 2×2 minors. The first equals $mJg(m + \sigma/(1+J)^2) > 0$. In the other two, both products are positive, because the diagonal entries are negative and the pairs $(\mathcal{J}_{13}, \mathcal{J}_{31})$ and $(\mathcal{J}_{23}, \mathcal{J}_{32})$ have opposite signs. In particular $a_2 \geq -\mathcal{J}_{13}\mathcal{J}_{31} \geq \alpha_{\max} H s' \gamma_A$.

The third condition. With $a_1 > m$ from Lemma 5.9,

$$a_1 a_2 > m \gamma_A \alpha_{\max} H s' > m \gamma_A \alpha_{\max} g H s' = a_3,$$

because $g = H/(H+v) < 1$. So $a_1 > 0$, $a_3 > 0$ and $a_1 a_2 > a_3$, the Routh–Hurwitz conditions for a cubic. $\square$

The two row operations are the energy budget of Section 5.3, read off the Jacobian: the new first row is the gradient of $\dot{H} + \dot{F}$, where recruitment cancels. The new third row is the gradient of $\dot{W}_\kappa$, where the brood cancels.

Two consequences follow. No positive equilibrium of (M) can lose stability through a Hopf bifurcation, in which a pair of complex conjugate eigenvalues crosses the imaginary axis and periodic orbits branch off [24, Ch. 3]. For a cubic with $a_1, a_3 > 0$ a pair of purely imaginary roots $\pm i\omega$ requires $a_1 a_2 = a_3$, which (5.15) excludes. No oscillating colonies therefore branch off from the positive equilibria. And the loss of the positive equilibrium at the existence boundary is not a stability failure. Since $H/(H+v) \approx H/v$ for small H , (5.15) gives $a_3 = O(H^2)$ and a_2 vanishes linearly with H^* , so as $p \downarrow p^*$ two eigenvalues tend to zero while the third stays near -0.587 . Remark 5.28 explains the pair.

5.2.3 Relaxation times

Local stability says a perturbed colony returns. It does not say when. The slowest eigenvalue does and the answer matters here more than usual, because the hornet season is short.

Remark 5.11 (Critical slowing down at both boundaries). The relaxation time is smallest in the middle of the band, about 124 d near $p \approx 0.49$ ($V \approx 6.6$) on a dense scan and grows without bound at p^* and at p_{acc} . At the lower boundary this is the pair of vanishing eigenvalues above. At the upper boundary the equilibrium escapes to $f = \infty$ and the third column of (5.12), which carries the factor $s'(f^*)$, tends to zero with it.

Every relaxation time in Table 5.1 is comparable to or longer than the hornet season and Section 6.4 draws the consequence.

Table 5.1: Relaxation time $1/|\operatorname{Re} \lambda_{\text{slow}}|$ across the existence band at $m = m_0$. It grows without bound towards both boundaries (Remark 5.11). Last column: the same for the rightmost characteristic root of the linearised delay model (K), which is real at every point, with no root in $\operatorname{Re} \lambda \geq 0$ (numerical control, Appendix B.2). A hornet season lasts 70–150 d.

p	V [hornets]	$\operatorname{Re} \lambda_{\text{slow}} [\text{d}^{-1}]$	relaxation [d]	delay model [d]
0.4160	8.05	−0.000178	5 631	9 576
0.4300	7.74	−0.002826	354	569
0.4500	7.33	−0.005673	176	263
0.4800	6.73	−0.007947	126	169
0.5100	6.18	−0.004103	244	267
0.5170	6.05	−0.000206	4 846	4 863

5.3 Global behaviour

Local stability leaves the main question open. A colony that has been knocked down does not start near its equilibrium. It may have had a bad winter, thrown a swarm or spent a month under hornets. Whether it returns is a global question and in three dimensions the planar tools are unavailable. What is available is better suited anyway: the model carries an exact budget identity, the analogue of the conservation principle of a chemostat.

5.3.1 The energy budget

The identity is built on one constant, $\kappa = \gamma_B/\varphi = 0.1620$ g: a consumption rate times the mean time in the brood compartment, that is, the honey one bee eats before she pupates. It is Harbo's 0.163 g per reared worker up to rounding, by construction of γ_B (Section 3.2.2). It is the weight with which the adults enter the function W_κ of Lemma 5.12, which for $p < p_\kappa$ decreases along every solution and so serves as a Lyapunov function, and it gives this weight a physical reading.

Lemma 5.12 (Energy budget). *Let $W_\kappa := f + \kappa(H + F)$ and*

$$p_\kappa := \frac{\gamma_A + \kappa m}{c} = 0.319480 \quad (V = 10.47 \text{ hornets at } m_0) \quad (5.16)$$

Then along every solution of (M) in Ω ,

$$\dot{W}_\kappa = cF(p - p_\kappa) - \gamma_A H \quad (5.17)$$

In particular, if $p < p_\kappa$ then $\dot{W}_\kappa < 0$ at every point of Ω with $(H, F) \neq (0, 0)$ and $\dot{W}_\kappa = 0$ exactly on the food axis $\{H = F = 0\}$.

Proof. Insert (M). The recruitment terms $\pm HR$ cancel in $\dot{H} + \dot{F}$ and the brood terms cancel between κLS in $\kappa(\dot{H} + \dot{F})$ and $-\kappa LS$ in $\dot{f}$:

$$\dot{W}_\kappa = cpF - \gamma_A(H + F) - \kappa LS + \kappa(LS - mF) = F(cp - \gamma_A - \kappa m) - \gamma_A H$$

For $p < p_\kappa$ both terms are ≤ 0 and both vanish only if $F = 0$ and $H = 0$. $\square$

Remark 5.13 (The break-even condition read as a balance sheet). W_κ is the colony's total energy capital measured in honey: the stores in the comb, plus the honey κ already invested in each standing adult. Equation (5.17) is then a per-forager balance sheet,

$$\underbrace{cp}_{\text{gross intake}} - \underbrace{\gamma_A}_{\text{own upkeep}} - \underbrace{\kappa m}_{\text{replacement cost}} \leq 0,$$

and the *break-even condition* $p > p_\kappa$ reads: *a forager must bring in enough to cover her own maintenance and the cost of rearing her own replacement.* Neither L , $\alpha_{\min}$, $\alpha_{\max}$, σ , b nor

v appears. The recruitment parameters drop out because recruitment moves bees between compartments without creating or destroying energy. The constants L and v drop out because the weight κ makes the brood terms cancel.

The price is that the condition charges the forager with none of the hive bees' upkeep: the term $-\gamma_A H$ in (5.17) is simply discarded. The condition is therefore not sufficient and not the load at which the positive equilibrium disappears (Section 5.5 charges the hive bees too). It is *necessary* for persistence in the sense of Definition 5.1, because $p_\kappa < p_E$ (Theorem 5.17, Proposition 5.19).

Origin of the budget identity. Its bookkeeping is that of a chemostat, whose conserved quantity $S+x/Y$ [47] counts substrate and biomass in one currency: the stores play the substrate, the adults the biomass and κ the yield coefficient. The analogy is one of bookkeeping, not of structure: here the foragers collect the substrate and the cancellation of the sign-changing recruitment term has no chemostat counterpart.

The identity (5.17) also holds in the published delay model (K), once the pupae in capped cells are counted as invested capital (Remark D.1 in Appendix D.7).

5.3.2 A larger weight

Lemma 5.12 proves collapse only for $p < p_\kappa$, that is above 10.47 hornets, far beyond the existence boundary at 8.07. The reason is that the weight κ discards the term $-\gamma_A H$, the upkeep of the hive bees. This subsection determines the admissible range of the weight a in $W_a = f + a(H + F)$, that is, the weights for which $\dot{W}_a < 0$ still follows once p is small enough (Lemma 5.14). A larger admissible weight charges each forager with part of that upkeep and so proves collapse on a larger range of p . The largest one, $a_\star$, gives the collapse bound p_E (Remark 5.15).

Lemma 5.14 (The admissible range of the weight). *Let $W_a := f + a(H + F)$. Then along every solution of (M) in Ω*

$$\dot{W}_a = LS(a - \kappa) + F(cp - \gamma_A - am) - \gamma_A H, \quad (5.18)$$

and for every a with $0 < a \leq a_\star$, where

$$a_\star := \kappa + \frac{\gamma_A v}{L} = 0.1795 \text{ g}, \quad (5.19)$$

the first and last terms together satisfy

$$LS(a - \kappa) - \gamma_A H \leq -\gamma_A \frac{H^2}{H + v} \leq 0 \quad \text{on } \Omega \quad (5.20)$$

The bound fails for every $a > a_\star$. Consequently, if $p < p_E(a) := (\gamma_A + am)/c$, then $\dot{W}_a < 0$ at every point of Ω with $(H, F) \neq (0, 0)$.

Proof. Equation (5.18) follows by inserting (M) as in the proof of Lemma 5.12. The brood terms now leave $LS(a - \kappa)$. For $a \leq \kappa$ this term is nonpositive and (5.20) holds trivially, since $-\gamma_A H \leq -\gamma_A H^2/(H + v)$. For $\kappa < a \leq a_\star$ use $s(f) \leq 1$:

$$LS(a - \kappa) = Ls(f) \frac{H}{H + v} (a - \kappa) \leq L \frac{H}{H + v} \cdot \frac{\gamma_A v}{L} = \gamma_A v \frac{H}{H + v},$$

so

$$LS(a - \kappa) - \gamma_A H \leq \gamma_A H \left(\frac{v}{H + v} - 1 \right) = -\gamma_A \frac{H^2}{H + v}$$

For $a > a_\star$ take $f \rightarrow \infty$ (so $s \rightarrow 1$) and $H \rightarrow 0^+$: the left-hand side of (5.20) behaves like $H[L(a - \kappa)/v - \gamma_A] > 0$, so the bound fails. Finally, with (5.20),

$$\dot{W}_a \leq -\gamma_A \frac{H^2}{H + v} - c(p_E(a) - p)F, \quad (5.21)$$

and for $p < p_E(a)$ both terms are ≤ 0 and vanish together only at $H = F = 0$. $\square$

Remark 5.15 (The collapse bound and the meaning of a_*). $p_E(a)$ is strictly increasing in a , so the largest admissible weight proves collapse over the largest range of p . It gives the *collapse bound*

$$p_E := p_E(a_*) = \frac{\gamma_A + a_* m}{c} = 0.346430, \quad V_E = 9.73 \text{ hornets} \quad (5.22)$$

The difference $a_* - \kappa = \gamma_A v / L = 0.0175$ g has a reading of its own. At low colony size a hive bee rears at most $L/v = 0.4$ bees a day, since $LS \leq (L/v)H$. Raising one adult therefore ties up at least $v/L = 2.5$ nurse-days and each nurse-day costs γ_A grams of honey. So a_* is the full honey cost of one adult: the food she eats as a larva, plus the upkeep of the nurses for the time it takes to rear her, at the most efficient nursing rate the model allows. Per forager and at replacement rate m , that is mv/L nurses, 0.385 at m_0 . The gap to $V^* = 8.07$ comes from bounding $s(f) \leq 1$, (5.31). The interval $p_E < p < p^*$, that is $V^* < V < V_E$, in which no positive equilibrium exists but collapse is not proved in general, is called the *window* below (Section 5.5).

Remark 5.16 (Other weights). Separate weights $W = f + aH + b_*F$ leave the sign-indefinite term $HR(b_* - a)$ and gain only two per cent in a numerical search ($p \approx 0.354$, $V \approx 9.5$), so the collapse bound is sharp within the family of equal weights but not overall.

5.3.3 Collapse for $p < p_E$

Theorem 5.17 (Global collapse). *Assume $p < p_E = (\gamma_A + a_* m)/c = 0.346430$, that is $V > 9.73$ hornets. Then for every solution of (M) that remains in Ω , $H(t) \rightarrow 0$, $F(t) \rightarrow 0$ and $f(t) \rightarrow f_\infty \geq 0$. In particular (M) has no positive equilibrium and no periodic orbit in this regime.*

Proof. Write $W := W_{a_*} = f + a_*(H + F)$. Since $p < p_E = p_E(a_*)$, (5.21) holds along the solution: $\dot{W} \leq -\gamma_A H^2/(H + v) - c(p_E - p)F \leq 0$.

Boundedness. $W \geq 0$ on Ω and W is nonincreasing, so $f \leq W(0)$ and $H, F \leq W(0)/a_*$ for all $t \geq 0$.

Integrability. Integrating (5.21) and using $W \geq 0$,

$$\int_0^\infty F(t) dt \leq \frac{W(0)}{c(p_E - p)}, \quad \int_0^\infty \frac{\gamma_A H^2}{H + v} dt \leq W(0) \quad (5.23)$$

Convergence. On the bounded solution the right-hand side of (M) is bounded, so F and $H^2/(H + v)$ have bounded derivatives and are uniformly continuous. Barbalat's lemma [20, Lemma 8.2] gives $F \rightarrow 0$ and $H^2/(H + v) \rightarrow 0$, hence $H \rightarrow 0$. Finally W is nonincreasing and bounded below, so $W \rightarrow W_\infty$ and $f = W - a_*(H + F) \rightarrow W_\infty =: f_\infty \geq 0$.

A positive equilibrium or a periodic orbit remains in Ω , so $H, F \rightarrow 0$ along it, which forces $H = F = 0$ throughout, where every point is an equilibrium. $\square$

LaSalle's principle [20, Thm. 4.4] would give the same limit but asks for a compact positively invariant set, which a sublevel set of W need not be, since other solutions may leave Ω through $\{f = 0\}$. Barbalat's lemma needs only the one solution and carries over to the delay model. The theorem fixes the direction and not the date. The first bound in (5.23) is the number of forager-days the colony's capital can still pay for, whatever it does with its bees. It diverges as $p \uparrow p_E$ and the time covered can be long: at $V = 9.72$, at the collapse bound, the tested colonies take 480 to 680 days to die (Table B.7), against a season of 70 to 150 days.

Corollary 5.18. *Assume $p < p_E$. Every solution of (M) that starts in Ω does exactly one of two things. Either it remains in Ω for all $t \geq 0$ and then $H, F \rightarrow 0$ by Theorem 5.17. Or it reaches the face $\{f = 0\}$ at a time T with $H(T) + F(T) > 0$ and leaves Ω there: the stores are exhausted while bees are still alive and from then on the model no longer describes a colony.*

Proof. The faces $\{H = 0\}$ and $\{F = 0\}$ cannot be crossed outwards (Lemma D.2), so a solution that leaves Ω does so through $\{f = 0\}$. A point of the food axis is an equilibrium, so a solution

that reaches $\{f = 0\}$ with $H + F = 0$ stays there and does not leave. Hence $H(T) + F(T) > 0$ at the exit time. $\square$

The second alternative is a gap in the model, not in the argument: (M), like Khoury's equations, does not say what a colony with an empty comb does. The natural rule is that it eats what comes in. That rule closes the gap.

Proposition 5.19 (Collapse with rationing at the empty comb). *Close (M) at $\{f = 0\}$ by rationing: where $f = 0$ and the right-hand side of the f -equation is negative, set $\dot{f} = 0$, so that the colony consumes exactly its intake. Everywhere else (M) is unchanged. For this closed system every point of Ω has a unique solution for all $t \geq 0$ [12]. If $p < p_E$, every solution of this system collapses and $f(t) \rightarrow f_\infty \geq 0$.*

Proof. On the face, $s(0) = 0$ gives $LS = 0$, so there $\dot{W} = a_\star(LS - mF) = -a_\star mF$. Where the rule is inactive, (5.21) holds. Hence $\dot{W} \leq -\eta F$ almost everywhere, with $\eta := \min\{c(p_E - p), a_\star m\} > 0$. As in Theorem 5.17, $W \geq 0$ is nonincreasing, the solution is bounded, $\int_0^\infty F dt \leq W(0)/\eta$ and since the H - and F -equations are unchanged, $\dot{F}$ is bounded and Barbalat's lemma gives $F \rightarrow 0$. If $H(0) = 0$, then $H \equiv 0$ and F decays at rate m . Otherwise the u -equation, unchanged as well, gives $u \geq u_-/2$ from some time on (Lemma 5.23) and $H + F = F/u \rightarrow 0$. Finally W converges and with it f . $\square$

Under this rule the second alternative of Corollary 5.18 disappears and since the rule changes nothing inside Ω , every other result stands. An exit is not starvation by itself: it occurs from suitable states at any load. At $V = 0$ the rationed colony recovers (Remark 5.21).

The collapse does not depend on the reduction of Section 4.4.

Theorem 5.20 (Collapse in the published delay model). *Consider (K) with the substitutions of Section 4.1, a continuous non-negative initial history of B on $[-\tau, 0]$ and $H(0), F(0), f(0) \geq 0$. Assume $p < p_E = (\gamma_A + a_\star m)/c$. Then every solution with $f(t) \geq 0$ for all $t \geq 0$ satisfies $B, H, F \rightarrow 0$ and $f \rightarrow f_\infty \geq 0$.*

The proof weights pupae and brood as well, $W_D := f + a_\star(H + F + \Pi) + (a_\star - \kappa)B$ with Π as in Remark D.1, shows that $\dot{W}_D$ obeys the estimate of Lemma 5.14 and concludes with Barbalat's lemma as above. It is given in Appendix D.7.

5.3.4 End states of collapse

Every collapsing solution ends at a point $(0, 0, f_\infty)$ of the food axis, which is a line of equilibria (Section 5.4). The possible end states therefore form a continuum and not a finite set, and Theorem 5.17 gives only $f_\infty \geq 0$. Where on the axis a colony ends is numerical and depends on the load. Integrated from the healthy state (9000, 2700, 1300), the model colony dies with food left in the comb at every tested $p > 0.1157$, that is below $V = 19.8$ hornets. The residual food rises with p , from zero at $p = 0.1157$ through 216 g at $p = 0.25$ to 312.3 g, just under f_{crit} , inside the window (Tables B.6 and B.7). It hardly depends on where the colony starts: at $p = 0.25$ the four initial states A to D of Appendix B.4, which span two orders of magnitude in population, leave 216.0 to 239.3 g. Section 5.4 explains why every such value lies below f_{crit} . Above $V = 19.8$ the stores run out first, with bees alive. The solution of (M) then leaves Ω (Figure 5.2). With rationing it stays on the empty comb and collapses there, at the end point $(0, 0, 0)$ of the axis. The two types of end state are one mechanism. (5.17) counts stores and bees as one capital and under pressure the only question is which component runs out first. A colony that still forages a little spends its honey slowly, while one that barely forages spends it at the rate $\gamma_A(H + F)$ of a population that has not yet had time to die.

Remark 5.21 (The exit through $\{f = 0\}$ is a property of the state). The load of about twenty hornets belongs to the healthy state and not to the model. A colony with many hive bees, few foragers and almost empty combs exhausts its stores at any load, because its foragers cannot cover the adults' upkeep before recruitment has caught up. From $(9000, 100, 5)$ the stores reach zero within about 0.1 days at $V = 7.3, 9.7$ and 11.0 hornets and from $(20000, 50, 1)$ even at $V = 0$ (numerical). With rationing (Proposition 5.19) these colonies spend 0.14 to 0.6 days on the empty comb. The one at $V = 0$ and the one at 7.3 hornets then recover, while the ones at 9.7 and 11.0 collapse inside the model. An exit alone is therefore not evidence of hornet damage.

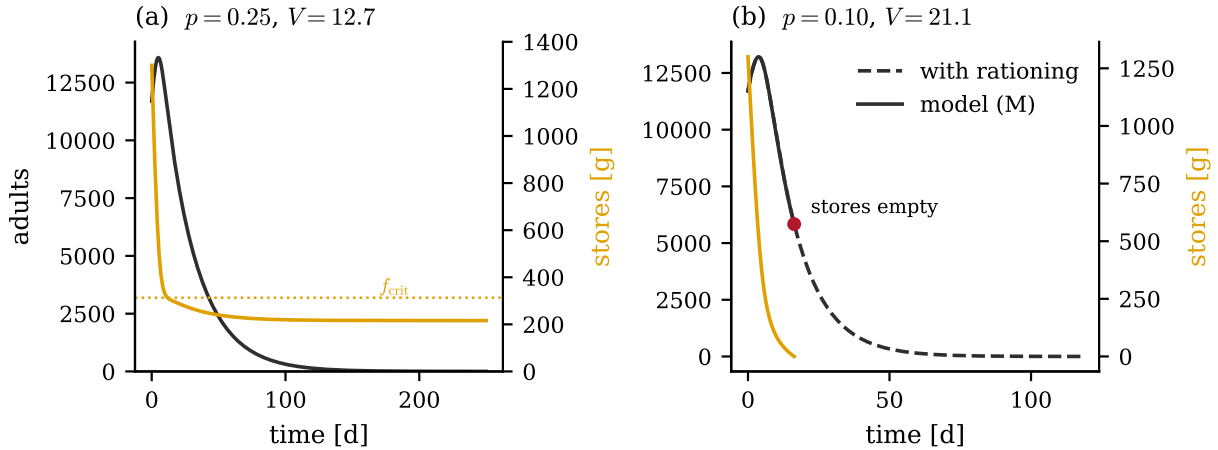


Figure 5.2: The two types of end state of Theorem 5.17, from the healthy state $(9000, 2700, 1300)$ at $m = m_0$. Every collapsing colony ends at a point $(0, 0, f_\infty)$ of the food axis, a line of equilibria, and f_∞ depends on p and on the initial state. (a) $p = 0.25$ ($V = 12.7$): the colony dies and leaves 216 g, below f_{crit} (dotted), as Proposition 5.31 requires. (b) $p = 0.10$ ($V = 21.1$): the stores run out after 16 days with about 5800 bees alive. The solution then leaves Ω . With rationing (Proposition 5.19) the colony falls below one bee after 118 days and ends at $f_\infty = 0$. LSODA, $\text{rtol} = 10^{-10}$ (Appendix B).

5.4 The food axis

The existence boundary of Section 5.1.2 is where the branch of positive equilibria runs into the food axis. What kind of loss that is depends on the stability of the axis itself. The same analysis shows which colonies below 9.73 hornets fall into the extinction set anyway: there is a floor for small colonies, measured in grams of honey. The positive equilibrium does not attract everything.

5.4.1 Degeneracy of the food axis

The food axis $\mathcal{E} = \{(0, 0, f_0) : f_0 \geq 0\}$ is a line of equilibria. Two obstacles stand in the way of the usual analysis. No point of $\mathcal{E}$ is hyperbolic, since the whole line consists of equilibria and there is a zero eigenvalue along it. And in the coordinates (H, F, f) the vector field is not continuously differentiable there. Because R depends on $F/(H + F)$, which has no limit at $H = F = 0$, entries of (5.12) such as $\sigma H^2/(H + F)^2$ stay bounded but have no limit on the axis: their value depends on the direction from which (H, F) approaches $(0, 0)$. Linearisation therefore does not apply.

Both dissolve for the same reason. Near the axis the (H, F) -block of (M) is asymptotically homogeneous of degree one, since $H/(H + v) \approx H/v$, so it splits into a size and a shape.

Proposition 5.22 (Homogeneity). *Let $N = H + F$, $u = F/N$, $s = s(f)$ and*

$$A_N(u, f) := \frac{L s(f)}{(1 - u)N + v} \quad (5.24)$$

Then, exactly on Ω with $N > 0$,

$$\frac{\dot{N}}{N} = A_N (1 - u) - m u, \quad (5.25)$$

$$\dot{u} = (1 - u)[\alpha(f) - u(\sigma + A_N + m)] \quad (5.26)$$

In the limit $N \rightarrow 0$ at fixed f , $A_N \rightarrow A(f) := (L/v)s(f)$ and both right-hand sides lose their dependence on N .

The computation is elementary and is given in Appendix D.2.

Since $A_N > 0$ for every $N \geq 0$, the bracket in (5.26) is strictly decreasing in u at fixed N and f , so the interior zero of (5.26) is unique and attracting for every $u_0 \in [0, 1)$. Near the axis, where $A_N \rightarrow A$, it is

$$u^*(f) = \frac{\alpha(f)}{\sigma + A(f) + m}, \quad (5.27)$$

and it depends on f alone. So near the axis the age structure settles first, at the rate $(1 - u^*)(\sigma + A + m)$, about 0.59 d^{-1} at f_{crit} . Equation (5.25) then gives a well-defined transverse growth rate. Away from the axis (5.26) still gives a global statement.

Lemma 5.23 (The age structure is confined to a band). *Let $u_- := \alpha_{\min}/(\sigma + L/v + m)$ and $u^+ := (\alpha_{\min} + \alpha_{\max})/(\sigma + m)$ (0.192 and 0.553 at m_0 , that is $0.81 \leq H/F \leq 4.2$). Along every solution with $H(0) > 0$ the interval $[u_-, u^+]$ is positively invariant and u enters every neighbourhood of it in finite time.*

Proof. $H(t) > 0$ for all t , since $\{H = 0\}$ is invariant, so $1 - u > 0$. By (5.26), with $\alpha \geq \alpha_{\min}$ and $0 \leq A_N \leq L/v$,

$$(1 - u)[\alpha_{\min} - u(\sigma + L/v + m)] \leq \dot{u} \leq (1 - u)[\alpha_{\min} + \alpha_{\max} - u(\sigma + m)]$$

The left side is positive for $u < u_-$ and the right side negative for $u > u^+$, so every interval $[u_- - \varepsilon, u^+ + \varepsilon]$ is positively invariant and so is their intersection. For $u \leq u_- - \varepsilon$ the left side is at least $(1 - u_-)(\sigma + L/v + m)\varepsilon$. For $u \geq u^+ + \varepsilon$, $w := 1 - u$ satisfies $\dot{w} \geq (\sigma + m)\varepsilon w$ and grows exponentially. $\square$

The ratio need not settle at a particular value. It enters a band of fixed width and stays there, whatever the stores and the load. That alone bounds the population.

Corollary 5.24 (The population is bounded at every load). *Along every solution with $H(0) > 0$ that remains in Ω , $\limsup_{t \rightarrow \infty} N(t) \leq L/(m u_-)$, about 68 000 bees at m_0 , for every p . The stores grow at most linearly, since $\dot{f} \leq c p N$.*

Proof. $\dot{N} = LS - mF \leq L - m u N$. By Lemma 5.23, $u \geq u_- - \varepsilon$ from some time on, so $\limsup N \leq L/(m(u_- - \varepsilon))$ for every $\varepsilon > 0$. $\square$

5.4.2 The transverse growth rate and f_{crit}

Lemma 5.25 (Transverse growth rate). *Assume $L/v > \alpha_{\min}$ (at the reference values $0.4 > 0.25$). Let*

$$\lambda(f) := A(f)(1 - u^*(f)) - m u^*(f) \quad (5.28)$$

Then λ is strictly increasing on $(0, \infty)$ and $\lambda(f) = 0$ is equation (5.11). Its unique zero is therefore the critical store level f_{crit} of Proposition 5.6. In this sense, with $\lambda(f_0)$ the growth rate of N near $(0, 0, f_0)$ once u has settled, the point $(0, 0, f_0)$ of the food axis is transversally attracting for $f_0 < f_{\text{crit}}$ and transversally repelling for $f_0 > f_{\text{crit}}$. Theorem 5.26 turns this into a stability statement.

The proof is algebra and one elementary estimate. It is given in Appendix D.3. Monotonicity needs no extra hypothesis: it follows from $\sigma > \alpha_{\min} + \alpha_{\max}$, the condition of Section 3.1.2 under which recruitment changes sign at every food level. Measured in the full nonlinear system near the axis, the growth rate of N agrees with $\lambda(f)$ to six decimals (Table B.5).

Lemma 5.25 gives the sign of a growth rate at a *fixed* food level. Because f moves while N decays, a stability statement needs more and the next theorem supplies it.

Theorem 5.26 (Stability of the food axis). *Let $f_0 > 0$ with $f_0 \neq f_{\text{crit}}$.*

- (i) *If $f_0 < f_{\text{crit}}$, the equilibrium $(0, 0, f_0)$ is Lyapunov stable and nearby solutions converge to nearby points of the axis: for every $\eta > 0$ there is $\rho > 0$ such that every solution with $N(0) < \rho$ and $|f(0) - f_0| < \rho$, whatever its shape $u(0)$, satisfies $N(t) < \eta$ and $|f(t) - f_0| < \eta$ for all $t \geq 0$, $N(t) \rightarrow 0$ exponentially and $f(t) \rightarrow f_\infty$ with $|f_\infty - f_0| < \eta$.*
- (ii) *If $f_0 > f_{\text{crit}}$, the equilibrium $(0, 0, f_0)$ is unstable: there are solutions starting arbitrarily close to it along which N leaves a fixed neighbourhood.*

The proof (Appendix D.4) never tracks the shape exactly. By (5.25) the growth rate of N decreases in u , so decay needs only a lower bound on u and growth only an upper bound. Each comes from a barrier on which $\dot{u}$ has a fixed sign. For initial shapes close to $u^*(f_0)$, part (i) is an instance of the standard result on normally hyperbolic attracting manifolds of equilibria [1]. No single point of the axis attracts, since every point is an equilibrium. What attracts is the segment $f < f_{\text{crit}}$.

Proposition 5.27 (The branch ends where the axis changes stability). *Fix $m > 0$ and let $p \downarrow p^*(m)$ along the branch of positive equilibria of Proposition 5.5. Then the branch converges to $(0, 0, f_{\text{crit}}(m))$, the point of the food axis at which λ changes sign. The collision point is the same whether the boundary is reached through p or through m .*

Proof. H^* and $F^* = JH^*$ tend to zero and $f^* \rightarrow f_{\text{crit}}$ by Proposition 5.6. By Lemma 5.25, $\lambda(f) = 0$ is (5.11), the equation that fixes $s(f^*)$ at the boundary, so $\lambda(f_{\text{crit}}) = 0$ and λ changes sign there because it is strictly increasing. Since f_{crit} does not depend on p , neither does the collision point. $\square$

With Theorems 5.26 and 5.10 this gives the local picture except for one piece: the phase portrait at the collision point itself is not derived.

Remark 5.28 (The existence boundary as a store drift). In (N, u, f) the size and shape obey (5.25) and (5.26). The stores obey $\dot{f} = NG_3$ with $G_3 := cpu - \gamma_A - \kappa A_N(1 - u)$, since $\gamma_B LS/\varphi = \kappa A_N(1 - u)N$. All three are smooth up to $N = 0$: the non-differentiability at the food axis belongs to the coordinates (H, F, f) , not to the dynamics. The axis becomes the plane $\{N = 0\}$ and its equilibria with $u < 1$ the curve $\Gamma := \{N = 0, u = u^*(f)\}$ (Figure 5.4). On Γ the linearisation is block triangular with eigenvalues $\lambda(f)$, $-(1 - u^*)(\sigma + A(f) + m)$ and 0, so Γ changes from attracting to repelling at f_{crit} , where λ crosses zero.

The activity enters this local picture only through G_3 . At the critical point, with $u_c := u^*(f_{\text{crit}})$, $\lambda = 0$ gives $A(1 - u_c) = m u_c$ and $1/u_c = 1 + P_c$, so by (5.10)

$$G_3 = u_c[cpu - \gamma_A(1 + P_c) - \kappa m] = cu_c(p - p^*)$$

The existence boundary is the activity at which the stores of a colony at the critical point stop drifting: above it they rise near the axis, below it they fall. This also explains the two vanishing eigenvalues of Section 5.2: in the limit the positive equilibrium lies on Γ where λ vanishes, the direction along Γ gives the second zero and the third is the shape direction, $-(1 - u_c)(\sigma + A + m_0) = -0.587$.

A collision, not a fold. The stressor models of Booton et al. [5] and Dennis and Kemp [10] lose their colony at a fold: a stable and an unstable equilibrium meet and vanish and the colony jumps. Nothing of that kind happens here. The positive equilibrium stays stable up to the boundary, its population shrinks continuously to zero and it leaves the region of positive populations through the food axis, linearly in $p - p^*$: by (5.4) and (5.2), $dH^*/dp \geq Lcs(f^*)/(m\gamma_A) > 0$, since $s(f^*)$ increases with p . The Allee effect lives on the attracting part of the axis rather than in a saddle (Remark 5.30) and it comes from the food coupling: Dennis and Kemp [10] note that the eclosion function of Khoury et al. [21] does not by itself constitute an Allee effect. What looks sudden in the field has two other sources, the narrow band of about two hornets and the slowing down near its boundaries (Remark 5.11).

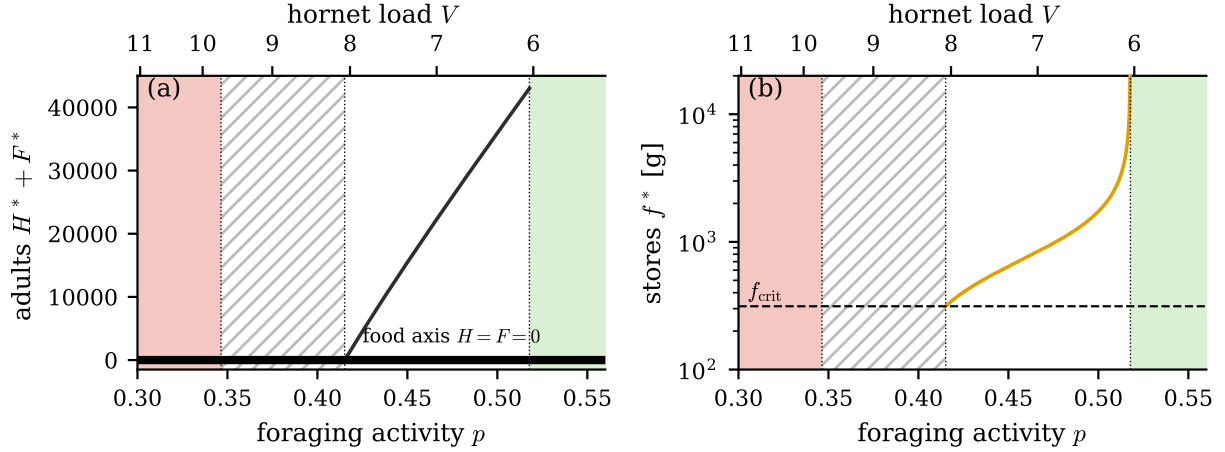


Figure 5.3: The branch of positive equilibria across the existence band at $m = m_0$ (Lemma 5.3), with the hornet load on the top axis. (a) The adults $H^* + F^*$ fall to zero at p^* , where the branch meets the food axis $H = F = 0$ (thick), a line of equilibria. (b) The stores f^* run to infinity at p_{acc} and down to f_{crit} (dashed) at p^* (Proposition 5.27). Shading: collapse proved left of p_E , window hatched, stores growing right of p_{acc} .

Remark 5.29 (Absent parameters). f_{crit} depends on $L, v, \alpha_{min}, \alpha_{max}, \sigma, b$ and m but on *none* of p, c, γ_A, γ_B . Only the food level enters, through the two behavioural switches it controls. Low stores mean a large $\alpha(f)$ and a small $s(f)$, so a small colony with low stores sends what few bees it has outside to die at rate m and raises too few replacements. The hornet does not need to be present for this. It only needs to have put the colony there once.

Remark 5.30 (Bistability and the role of f_{crit}). For every p in the existence band (5.7) the positive equilibrium is *not* globally attracting: by Theorem 5.26(i) every solution starting in the open set $\{0 < N < \rho, |f - f_0| < \rho\}$ around a point with $f_0 < f_{crit}$ converges to the axis, so the locally stable positive equilibrium (Theorem 5.10) coexists with an attracting segment of extinction states. How far its basin reaches is numerical (Figure 5.5): f_{crit} separates survival from collapse only for colonies close to extinction, larger colonies survive from stores well below it and from about 1 500 adults on even from nearly empty combs. Since both basins are open, a boundary solution that converges to the axis must end at f_{crit} , because Theorem 5.26(i) excludes $f_\infty < f_{crit}$ and Proposition 5.31 excludes $f_\infty > f_{crit}$. That is where the computed boundary ends as $N_0 \rightarrow 0$.

A collapsing colony cannot come to rest on the repelling part of the axis.

Proposition 5.31 (Limit point of a collapsing colony). *Let a solution of (M) with $H(0) > 0$ remain in Ω and converge to a point $(0, 0, f_\infty)$ of the food axis. Then $f_\infty \leq f_{crit}$.*

The proof is in Appendix D.5. If f_∞ were above f_{crit} , the shape would be pushed below $u^*(f_\infty)$ plus a small margin and the population would grow at a rate close to $\lambda(f_\infty) > 0$, which contradicts convergence to the axis. The hypothesis $H(0) > 0$ is needed, because a colony without

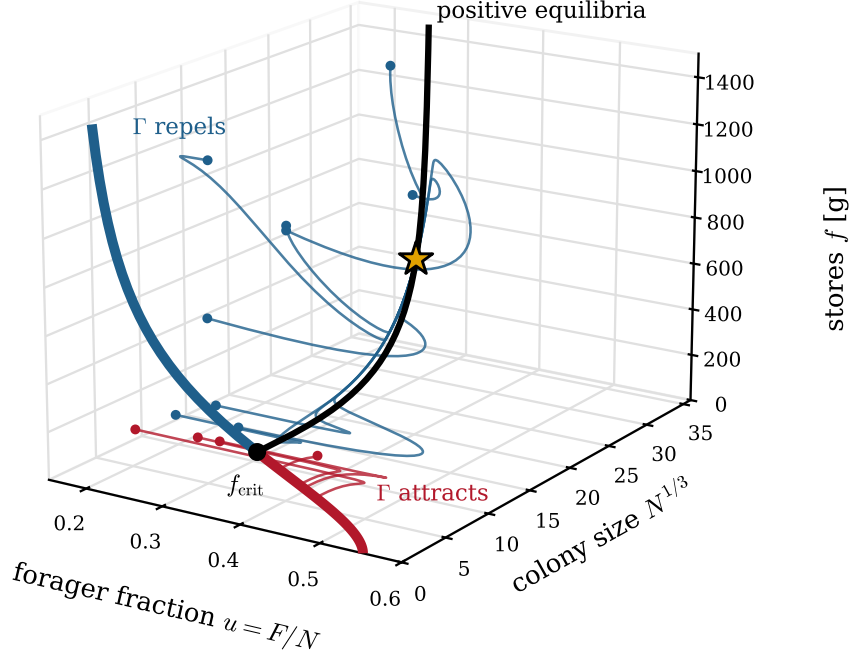


Figure 5.4: The collision at the existence boundary in the coordinates of Remark 5.28, size axis $N^{1/3}$. The food axis becomes the curve Γ , attracting below f_{crit} (red) and repelling above (blue), Theorem 5.26. Black: positive equilibria for $p^* < p \leq 0.495$, meeting Γ at $(u_c, 0, f_{\text{crit}})$ as $p \downarrow p^*$ (Proposition 5.27). Thin lines: solutions at $p = 0.45$, ending at the positive equilibrium (star) or on the attracting part of Γ .

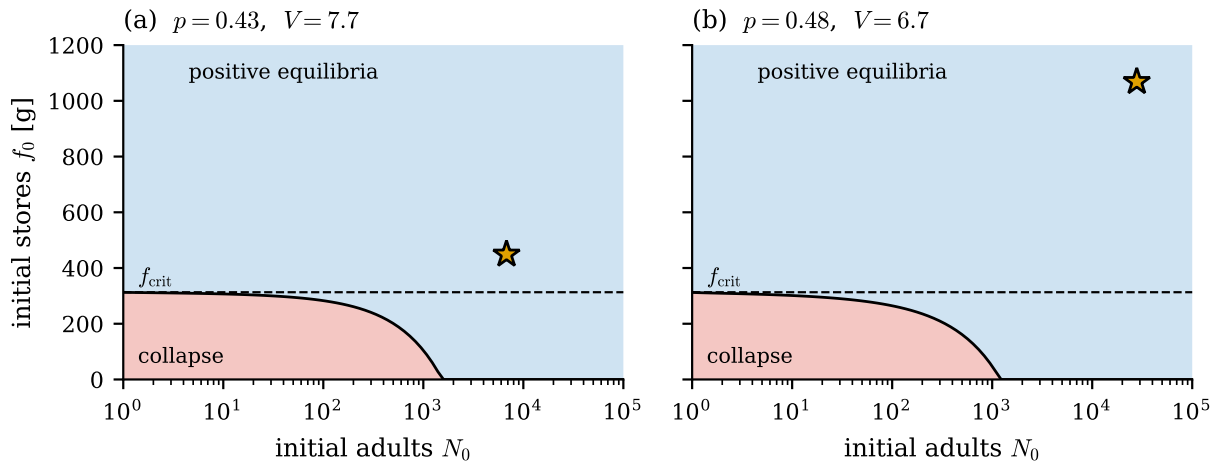


Figure 5.5: Basins of attraction inside the existence band (Remark 5.30), for initial states with $H_0:F_0 = 2:1$. Red: collapse ($H + F < 1$ within 4000 d, with rationing). Blue: convergence to the positive equilibrium (star). Dashed: f_{crit} . Boundary located by bisection in f_0 (LSODA, $\text{rtol} = 10^{-8}$).

hive bees loses its foragers at rate m whatever its stores. The residual food of Section 5.3.4 is consistent with it.

5.5 A common form for the four thresholds

Four quantities in the activity p have appeared: the accumulation boundary p_{acc} , the existence boundary p^* , the collapse bound p_E and the break-even bound p_κ . They answer different questions and have different origins. This section shows that they can be written in one bookkeeping form and that the real content lies in how the form is filled and in the order it forces. The closed form of p^* (Proposition 5.6) supplies the last piece.

The nurses at the critical store level. A forager dies at rate m , so her replacement must be reared at rate m . Near extinction a hive bee rears at most $(L/v)s(f)$ bees a day, so replacing one forager occupies $mv/(Ls(f))$ hive bees. At the collision the colony holds exactly these nurses and no other hive bees:

$$P_c = \frac{mv}{Ls^*} = \frac{1}{s^*} \cdot \frac{mv}{L} \quad (5.29)$$

These are the nurses that $a_\star$ charges (Remark 5.15), now at the brood survival s^* of the critical store level rather than at $s = 1$: 1.367 against 0.385 per forager at m_0 . Setting $p^*(m) = 1$ and solving for m gives $m^* = 0.4624$ of Corollary 5.8.

The common form. Write

$$p_X := \frac{\gamma_A(1+X) + \kappa m}{c} \quad (5.30)$$

It is the activity at which a forager's intake cp just covers her own upkeep γ_A , the honey κm her replacement eats as a larva and the upkeep $\gamma_A X$ of X hive bees. All four have this form, with different X (Table 5.2).

The form by itself is bookkeeping: any activity p can be written as p_X for a suitable X . The content lies in how X is fixed and there are two routes. For p^* and p_{acc} , (5.30) is the food balance at an equilibrium, step (i) of Lemma 5.3, evaluated at the ratio H/F that recruitment selects: at the critical store level for p^* , with full combs for p_{acc} . These are statements about where a positive equilibrium exists. For p_κ and p_E , (5.30) is where a Lyapunov estimate changes sign and X is what that estimate can absorb: nothing for the weight κ , the minimal nursing cost mv/L for $a_\star$. These are conditions for collapse and say nothing about equilibria directly. The one statement the form does make, beyond relabelling, is the order.

Table 5.2: The two bounds and two boundaries in the form (5.30) at $m = m_0$, where X is the number of hive bees whose upkeep is charged to one forager. The first two rows are sufficient for collapse (Lyapunov estimates), the last two are the boundaries of the existence band. Hornet loads with $\beta = 0.109$.

threshold	hive bees charged, X	X	p_X	V	status
p_κ , break-even	none	0	0.3195	10.47	collapse proved below
p_E , collapse bound	nurses needed to rear her replacement at full brood survival, mv/L	0.385	0.3464	9.73	collapse proved below
p^* , existence boundary	the same nurses at the brood survival of f_{crit} , $P_c = mv/(Ls^*)$	1.367	0.4151	8.07	positive equilibrium exists iff $p^* < p < p_{\text{acc}}$
p_{acc} , accumulation boundary	hive bees per forager with full combs, Q	2.833	0.5178	6.04	above it a well-stocked colony gains stores (numerical)

Proposition 5.32 (The order of boundaries and bounds). *For every $m > 0$ and all positive parameters with $\sigma > \alpha_{\min} + \alpha_{\max}$ and $L/v > \alpha_{\min}$,*

$$p_{\kappa} < p_E < p^* < p_{\text{acc}}$$

Proof. Since p_X is increasing in X , it suffices to order the four values of X . $0 < mv/L$ is clear and $mv/L < P_c = mv/(Ls^*)$ because $s^* < 1$ (Appendix D.3). For the last step, $J_c = 1/P_c$ and $J_Q = 1/Q$ are the positive roots of $mJ^2 + (m + \sigma - \alpha)J - \alpha = 0$ with $\alpha = \alpha(f_{\text{crit}})$ and $\alpha = \alpha_{\min}$ respectively (Proposition 5.6 and (5.5)). That root increases with α and $\alpha(f_{\text{crit}}) > \alpha_{\min}$ because $f_{\text{crit}} < \infty$. Hence $J_c > J_Q$, that is $P_c < Q$. $\square$

Consequences for the open window. Between p_E and p^* there is no equilibrium and the weight a_* does not prove collapse. The closed form says exactly why. The estimate behind a_* bounds the brood survival by its maximum, $s(f) \leq 1$, while at the collision the brood survives at s^* . The window is that factor and nothing else:

$$p^* - p_E = \frac{\gamma_A m v}{c L} \cdot \frac{1 - s^*}{s^*} = 0.0687, \quad \text{that is 1.66 hornets at } m_0 \quad (5.31)$$

On stores at or below f_{crit} the sharper bound $s(f) \leq s^*$ is available and there it closes the window.

Proposition 5.33 (Collapse in the window while the stores stay low). *Fix $m > 0$ and $p < p^*(m)$. Put*

$$a_c := \kappa + \frac{\gamma_A v}{L s^*}, \quad W_c := f + a_c(H + F)$$

Then on $\Omega \cap \{f \leq f_{\text{crit}}\}$

$$\dot{W}_c \leq -\gamma_A \frac{H^2}{H + v} - c(p^* - p)F \quad (5.32)$$

Consequently every solution of (M) that remains in Ω and satisfies $f(t) \leq f_{\text{crit}}$ for all $t \geq T$, for some $T \geq 0$, has $H(t) \rightarrow 0$, $F(t) \rightarrow 0$ and $f(t) \rightarrow f_{\infty} \leq f_{\text{crit}}$.

The proof runs the estimate of Lemma 5.14 with $s(f) \leq s^*$ in place of $s(f) \leq 1$ and then the argument of Theorem 5.17 on $[T, \infty)$. It is given in Appendix D.8. At m_0 , $a_c = 0.2241$ g against $a_* = 0.1795$ g. The larger weight fails the global bound of Lemma 5.14, which is why it is confined to low stores. A colony in the window can therefore escape collapse only by holding its stores above f_{crit} at arbitrarily late times. Every trajectory of Table B.7 ends below it. The window is reduced to one question, not closed (Section 7.4).

5.6 Biological interpretation

The cascade answers three biological questions in a fixed order: how many hive bees one forager can support (P , set by intake, upkeep and replacement cost), how full the combs must be for the colony to run at that ratio and, last, out of the brood-care bottleneck, how big the colony is. A stronger queen does not raise the sustainable ratio. She raises the size and every extra bee eats. Each term of (5.13) pulls a perturbed colony back and every positive equilibrium the model has is stable.

The break-even condition is a balance a beekeeper could check on the back of an envelope (Remark 5.13). If a forager collects less than her own upkeep and the cost of raising her replacement, the colony is spending capital and no amount of reorganisation inside the hive changes that. The recruitment machinery moves bees. It does not make honey.

Finally there is a floor for small colonies and it is measured in grams of honey rather than in bees. A small enough colony whose stores lie below roughly three hundred grams cannot climb

back out, because at that food level its own behavioural rules push bees outward and brood production down. How small is small enough the proof does not say. Numerically, at $p = 0.48$ about two hundred adults at 250 g still recover (Figure 5.5b). Two small colonies side by side under the same hornet pressure can therefore end up in different places: the one with fuller combs can recover, the one with emptier combs may not. A stable positive equilibrium means that a colony has somewhere to go, not that it gets there. That depends on where it starts.

Chapter 6

The two channels

By (4.5) the hornet adds at most $\nu h_0/(\zeta e) = 3.62 \cdot 10^{-4} \text{ d}^{-1}$ to the forager death rate, a quarter of a per cent of m_0 . Along the mortality axis at full foraging activity the existence boundary lies an increase of 0.31 d^{-1} away (Corollary 5.8). The two numbers belong to different activities, so they fix the scale of question Q3 rather than settle it. This chapter puts both channels in one plane, compares them like for like along the path the hornet actually traces and asks what the thresholds depend on.

6.1 The (p, m) plane and the hornet trajectory

The hornet enters (M) at two places and nowhere else, through p and through m . The results of Chapter 5 hold for arbitrary pairs (p, m) , so the object they describe is a two-parameter family of systems. Proposition 5.5 cuts its plane into three regions. Right of the accumulation boundary $p_{\text{acc}}(m)$ there is no positive equilibrium: in the integrations a well-stocked colony settles at the ratio Q with growing stores, a small one with stores below f_{crit} still dies. Inside the band a unique positive equilibrium exists and is stable (Theorem 5.10). Left of the existence boundary $p^*(m)$ the branch has met the food axis and left. Inside that third region lies a fourth: left of the line

$$p_E(m) = \frac{\gamma_A + a_\star m}{c}, \quad a_\star = 0.1795 \text{ g}, \quad (6.1)$$

Theorem 5.17 proves collapse for every solution that stays in the model's domain. Between $p_E(m)$ and $p^*(m)$ lies the window, where collapse is proved only while the stores stay at or below f_{crit} (Proposition 5.33).

The band is narrow and almost of constant width and it slides to the right as mortality grows (Table B.4, Appendix B.3). Its upper boundary reaches $p = 1$ at $m = 0.4010$ and its lower boundary at $m^* = 0.4624$. Above m^* no positive equilibrium exists at any activity.

The trajectory the hornet actually traces. A given hornet load fixes both parameters at once. As V runs from zero upwards, the colony moves along the curve

$$V \mapsto (p(V), m_0 + \mu(V)), \quad p(V) = e^{-\beta V}, \quad \mu(V) = \nu p(V) \text{ HF}(p(V)) \quad (6.2)$$

Over $V \in [0, 20]$, the range Requier et al. [37] observed, p falls from 1 to 0.113, while m rises from 0.154014 to 0.154362 at $V = 16.6$ and falls back to 0.154341, a total excursion of $3.5 \cdot 10^{-4}$. The path is parallel to the p -axis to within a quarter of a per cent of m_0 and crosses the three boundaries in turn. From $(p, m) = (1, m_0)$ the existence boundary is reached when activity falls by 58 % or when forager mortality triples. The decisive asymmetry is in how far each channel can push the colony: one covers its distance by eight hornets, the other covers about a thousandth of it and turns around.

6.2 Reach of the mortality channel

The two channels are not independent. The mortality channel is calibrated as a function of traffic, so switching paralysis off also switches off the rise in risk per flight. Predation acting

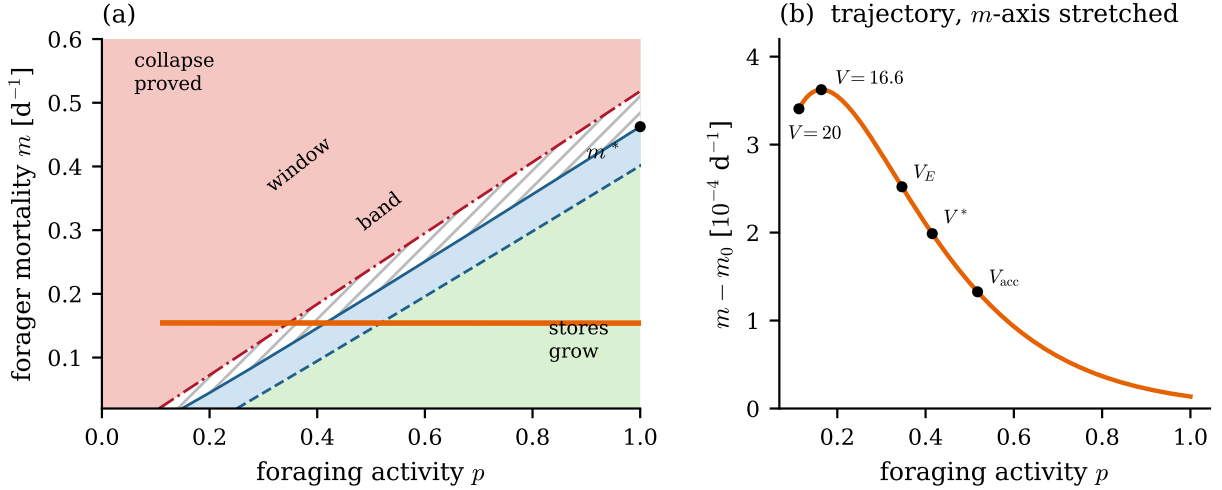


Figure 6.1: The (p, m) plane at the reference parameters. (a) The band of Proposition 5.5 (blue) carries the only positive equilibrium. Right of it there is no positive equilibrium and the stores grow. Left of the dash-dotted line $p_E(m)$ (6.1) collapse is proved (Theorem 5.17). Between it and the band lies the window (hatched), where collapse is proved only for low stores (Proposition 5.33). The hornet trajectory (6.2) is the horizontal line at $m \approx m_0$. (b) The trajectory for $0 \leq V \leq 20$, with the m -axis stretched about 1100-fold relative to (a). It turns back at $V = 16.6$, where μ is largest (Section 4.3.4).

alone is therefore not a scenario the calibration describes. The comparison is therefore made in three steps. The first two are like-for-like and the third is a counterfactual.

1. Shift of the existence boundary under paralysis. Carrying μ along the hornet trajectory moves p^* from 0.4151 to 0.4155, that is by 0.01 **hornets**. The other thresholds move by the same order (Remark 4.1). The captures recorded on video at the calibration colony run three to twelve times above the fitted curve if that colony held 3000 to 11000 foragers (Appendix A.2). Scaling HF by those factors moves the boundary by 0.03 to 0.10 hornets (numerical). Either way the shift is below the counting resolution of V .

2. Capture rate for a shift of one hornet. The required rate does not use the fitted form of HF, the scale of its activity axis or the way the load acts on it. By the closed form of Proposition 5.6, moving V^* down by one hornet needs $\Delta m = 0.025 \text{ d}^{-1}$ at the load $V^* - 1 = 7.07$, where a forager makes $\nu p = 1.39$ trips a day. That is a capture on 1.8% **of return flights**, one flight in fifty-six and nine times the highest rate in the fitted relation. The positive equilibrium at that load with $m = m_0$ holds about 6900 foragers, so the colony would have to lose about 170 bees a day to hornets, whereas the video colony lost about 13. Against the video count itself, for any forager population in the model's range, the observed rate is at most $4.2 \cdot 10^{-3} \text{ d}^{-1}$ (Appendix A.2), six times short of Δm . The target does not matter: for V_{acc} and V_E the same computation gives 1.75% and 1.9%.

3. Counterfactual: predation without paralysis. A colony at $(p, m) = (1, m_0)$ has no positive equilibrium to lose (Section 5.1.2). Along the mortality axis its stores stop accumulating at $m = 0.401$, a positive equilibrium exists between there and m^* and the existence boundary lies at

$$m^* = 0.4624 \text{ d}^{-1} = 3.00 m_0, \quad m^* - m_0 = 0.3084 \text{ d}^{-1}, \quad (6.3)$$

a capture on 10.3% of return flights at full traffic. The fitted relation attains at most $3.62 \cdot 10^{-4} \text{ d}^{-1}$ and only at $p = 0.164$, under heavy paralysis, so

$$\frac{m^* - m_0}{\max_V \mu} \approx \mathbf{851}, \quad (6.4)$$

and the video capture rates still fall short by a factor of 73 to 270. Even the nearer boundary at $m = 0.401$ is 680 times the bound away. The two sides of this ratio live at different activities, so it is a check on the scale of the channel rather than a statement about any load the hornet imposes. The bound is linear in νh_0 and does not depend on β . A channel that cannot reach the existence boundary cannot reach the collapse bound further out.

The factor splits into two. At full traffic and at the highest per-flight risk of the fitted relation, predation would add $\nu h_0 = 0.006 \text{ d}^{-1}$, still 51 times short: captures are rare. The fitted relation never attains that product, because its highest risk comes with the thinnest traffic. By (4.5) the peak of μ lies lower by the factor $\zeta e \approx 17$. The factor 851 is the product of the two.

This is the answer to Q3. The larger part is magnitude and step 2 shows it without the fitted form. The second part is behavioural: the risk per flight rises with the load while the number of flights falls, so although μ still rises up to 16.6 hornets, hence at every threshold of this thesis, the two factors are never large together. Only the behavioural channel carries the colony to the boundary at which its positive equilibrium disappears.

6.3 Sensitivity

β appears nowhere in (M). It enters only when a threshold in p is converted into hornets, so the thresholds in p are unaffected by the calibration of Section 4.3.2 and only their expression in V moves (Table 6.1). Thresholds are therefore stated in p first and in V second: a threshold in foraging activity is a property of the colony's energy budget, a threshold in hornets is that property seen through one fitted exponential. The trip rate ν enters only the bound (4.5), which is linear in it. At every published value under natural foraging, up to 24 trips a day, the mortality channel still falls short of (6.3) by more than two orders of magnitude (Table A.4).

Table 6.1: The three thresholds in hornets, $V = -\ln p/\beta$. The middle three rows span the digitisation spread of Table B.1 and the outer two, $\pm 20\%$ of the reference value $\beta = 0.109$, stand in for the wider statistical uncertainty of the fit. The thresholds in p are the same in every row.

β [hornet $^{-1}$]	V_{acc}	V^*	V_E
0.087 (−20 %)	7.56	10.10	12.18
0.107	6.15	8.22	9.91
0.109	6.04	8.07	9.73
0.120	5.48	7.33	8.83
0.131 (+20 %)	5.02	6.71	8.09

Remark 6.1 (Forage quality). In (M) the activity p occurs only in the term cpF , so at fixed m every threshold in p scales exactly as $1/c$. In hornets,

$$V_{\text{thr}}(c) = V_{\text{thr}}(c_0) + \frac{1}{\beta} \ln \frac{c}{c_0}, \quad (6.5)$$

the same shift for every threshold: a site that is 20% richer buys 1.7 hornets and a site twice as rich buys 6.4. The mortality increment μ does not rescale, since it depends on V through p and not through cp . The value $c_0 = 0.1$ is a plentiful-forage value (Table 4.1, (g)). At the October intake of Russell et al. [44], $0.45 c_0$, every threshold lies about 7 hornets lower, so under poorer autumn forage the answer to Q1 holds at still lower loads. At that intake the existence boundary lies at 0.7 hornets, the collapse bound at 2.4 and the accumulation boundary below zero, so the model colony loses stores in autumn even without hornets and the hornet deepens a deficit rather than removing a surplus. The distance $m^* - m_0$ on the mortality axis shrinks with c as well, from 851 to about 50 times the largest μ , so predation stays far out of reach at any forage.

The largest sensitivities. The intake c (Remark 6.1) and one modelling decision move the thresholds further than β and ν do. The model keeps the baseline mortality m_0 of a paralysed forager unchanged and scales only the per-flight predation risk with p (Section 4.2.1). The opposite extreme reads a bee in the nest as a bee at no risk and scales the whole forager mortality with flight activity,

$$m = p(V) [m_0 + \nu \text{HF}(p(V))], \quad (6.6)$$

which is close to what the BEEHAVE implementation of Requier et al. [37] does. A colony under (6.6) buys longevity with its paralysis: foragers accumulate, the population rises and the colony starves slowly instead of quickly, which is the sign Requier et al. [37] report (Section 7.2). Between the two extremes, let a fraction θ of m_0 be flight-related,

$$m(p) = m_0[(1 - \theta) + \theta p] + \mu,$$

so that $\theta = 0$ is the model of this thesis and $\theta = 1$ is (6.6). Table 6.2 gives the three thresholds, all of which depend on m . The two ends bracket every intermediate split, because $p^*(m)$, $p_E(m)$ and $p_{\text{acc}}(m)$ increase with m (Appendix D.6, with Q growing in m by (C.1)) and m decreases with θ at every $p < 1$. Each boundary is unique, since all three grow along $m(p)$ with slope below 0.4 in p .

Neither end is the correct reading of m_0 . A paralysed forager still ages and falls ill, which argues against $\theta = 1$. A forager's life is also spent on the wing and the hazard she faces there is plausibly a large part of m_0 , which argues against $\theta = 0$. Tracked foragers die at a roughly constant rate per unit time away from the colony, mostly before senescence, which points to the external risks of foraging rather than wear [52]. Their flightspans are approximately exponentially distributed [43]. No measurement in the calibration data fixes θ and Remark 7.1 says which one would. With respect to θ , the thresholds in hornets quoted in this thesis are the pessimistic end, $\theta = 0$ (with respect to forage they are not, Remark 6.1): the accumulation boundary lies between 6.0 and 10.9 hornets, the existence boundary between 8.1 and 16.9 and the collapse bound between 9.7 and 21.4. That bracket is wider than the whole $\pm 20\%$ envelope on β . It is of the same size as the forage caveat and points the other way, up to nine hornets upwards at V^* against about seven downwards for autumn forage. The robust content of the thresholds is therefore their order, their values in p and the comparison of the channels. The season loads of Section 6.4 are computed at $\theta = 0$ only. The answer to Q1 holds for $\theta \lesssim 0.95$, where V_E reaches twenty hornets. At $\theta = 1$ collapse is still proved under paralysis alone, but just beyond the observed loads, where $p(V)$ is extrapolated.

Table 6.2: V_{acc} , V^* and V_E when a fraction θ of the baseline mortality scales with flight activity, μ included. $\theta = 0$ reproduces Remark 4.1, $\theta = 1$ is (6.6). At $\theta = 1$, dropping the HF term (a change below 0.1 hornets) gives closed forms: $\tilde{p}_E = \gamma_A/(c - a_*m_0) = 0.0967$ (21.4 hornets) and $\tilde{p}_{\text{acc}} = \gamma_A(1 + \tilde{Q})/(c - \kappa m_0) = 0.3051$ (10.9 hornets), with $\tilde{Q}$ the positive root of $(\alpha_{\min} - k)Q^2 + (\alpha_{\min} - \sigma - 2k)Q - k = 0$, $k = \gamma_A m_0/(c - \kappa m_0)$.

θ	0	0.25	0.5	0.75	1
p_{acc}	0.5181	0.4785	0.4316	0.3753	0.3059
V_{acc} [hornets]	6.03	6.76	7.71	8.99	10.87
p^*	0.4155	0.3685	0.3128	0.2452	0.1590
V^* [hornets]	8.06	9.16	10.66	12.90	16.87
p_E	0.3469	0.2985	0.2423	0.1763	0.0975
V_E [hornets]	9.71	11.09	13.01	15.92	21.35

6.4 Field relevance

The relaxation time across the band is at least about 124 d in (M), longer in the delay model and diverges at both boundaries (Table 5.1). The season of heavy pressure lasts about 70 d,

Requier’s own window of loads above five hornets (Table A.5), to about 150 d, the more than five months of predation recorded by Monceau et al. [28]. The long-run state of the model is therefore not reached within a season. What can be computed over one season is the change in the stores. Table 6.3 integrates the model, with $m = m_0 + \mu(V)$, that is at $\theta = 0$, over 70 and 150 days from the four initial states A to D of Appendix B.4, which span two orders of magnitude in population. It locates the *season store-balance load*, at which the stores end the season where they started. Because the reduction of Section 4.4 is not a time-scale approximation, the table repeats the computation in the published delay model.

Table 6.3: Season store-balance load at $c = c_0$: the load V at which the stores after T days equal the initial stores. Below it the colony gains honey over the season and above it the colony loses honey. States A to D of Appendix B.4. Model (M): LSODA, $\text{rtol} = 10^{-9}$. Delay model (K): RK4, $\Delta t = 0.02$ d (unchanged at 0.01 d), constant brood history $B = LS(H_0, f_0)/\varphi$, so that both models start with equal fluxes. In every case the store change has exactly one sign change on $V \in \{0, 1, \dots, 20\}$, located by bisection. Numerical control.

season length	model	A	B	C	D
70 d	(M)	5.53	5.67	6.01	5.92
	delay	5.50	5.59	5.80	6.14
150 d	(M)	5.83	6.23	6.78	6.79
	delay	5.80	6.14	6.54	6.49

Across states, season lengths and both models the store-balance load lies between 5.5 and 6.8 hornets, the delay moves it by at most 0.3 hornets and the accumulation boundary $V_{\text{acc}} = 6.04$ lies inside that range. So V_{acc} is a closed-form proxy for the season store-balance load: over one season, the store balance of colonies of very different sizes changes sign within a hornet of it. The heuristic reason is the balance (5.30). A colony whose age ratio equals Q and whose brood production matches its losses has store drift $F[cp - \gamma_A(1 + Q) - \kappa m]$, which changes sign exactly at p_{acc} . The colonies of the table are near that state but not in it: Lemma 5.23 keeps their age structure in a band, not at Q . The spread in the table measures the distance. Both numbers belong to $\theta = 0$ and $c = c_0$. At $\theta = 1$ the closed-form proxy moves to 10.9 hornets (Table 6.2). At the October intake it lies below zero (Remark 6.1).

One study offers a comparison close enough to be worth making. Diéguez-Antón et al. [11] photographed hive entrances hourly for five to eleven months. Foraging decreased significantly above about five hornets in front of the entrance and broke down above ten. Colonies under loads above ten for long periods died. The five and ten hornets also serve as the predefined risk levels of their survival analysis. The accumulation and existence boundaries fall between those two figures. Their metric is close to Requier’s but not the same (Section 4.3.1), so the agreement is support from a different group using a different method, not a validation.

6.5 Biological interpretation

The earlier stressor models place their stressor on the mortality axis of this plane (Section 7.1). The hornet does not move a colony along that axis. It moves it along the other one and it moves it far. To reach the boundary, activity has to fall by more than half or mortality has to triple. Only the first of these is something the hornet can do.

The model has no winter, so it cannot predict one. What it can compute directly is the season’s store balance. Read biologically and at the pessimistic end $\theta = 0$, a colony above about six hornets ends the season with less honey than it started with. BEEHAVE, which models the winter, places the resulting losses months later (Section 7.2). Six hornets in front of an entrance is not a dramatic sight. Under plentiful forage it is enough to take away the surplus. Under autumn forage there is no surplus to take and the hornet deepens a deficit (Remark 6.1).

Chapter 7

Discussion

This chapter places the results in the literature, states the limitations of the model and its analysis, outlines further work and answers the three research questions.

7.1 Relation to previous work

Not new. Allee effects are well established in models of social bee colonies [5, 6, 10] and explicit Lyapunov functions in bee models exist [48]. Torres et al. [50] require that intake cover the consumption of larvae, hive bees and foragers, the closest analogue to the balances (5.30). Henry et al. [16] add pesticide-induced homing failure to the forager death rate of Khoury et al. [21]. Booton et al. [5] add a density-dependent stress mortality on both adult classes of the same model and obtain an Allee effect with a saddle-node bifurcation, the nearest predecessor, but with the stressor in mortality alone and no food compartment. Ratti et al. [36] model forager loss and homing failure explicitly, for *Varroa* and acute bee paralysis virus. They determine the critical loss rate by simulation, without a closed threshold or a global stability statement. Magal et al. [26] build a spatial model of collapse driven by “the fraction of contaminated forager bees that fail to return home”: channel 1 of this thesis, under pesticide contamination, in a different setting and with different mathematics. Each of these stressor models did one part of what is done here. A recent age-structured model fits pesticide effects to adult mortality, the laying rate and the half-saturation of egg survival [8]. A 2021 review contains no predator model [7]. The tools are standard: a Jacobian, Routh–Hurwitz, a budget identity, Barbalat’s lemma and comparison estimates.

New. Five contributions, each modest. To my knowledge and on a search of the literature citing Requier et al. [37] and Khoury et al. [22] up to September 2026, no earlier model gives explicit thresholds, in predator numbers, for a stressor that acts on foraging rather than on survival.

1. **The intake rate is tied to a measured predator load.** The substitution $c \mapsto cp(V)$ is calibrated against Requier et al. [37, Fig. 2a], so the thresholds come out in hornets, whereas Khoury et al. [22] vary c without an external referent.
2. **Both channels run at once and are compared in one plane** (Chapter 6), with the predation channel bounded at every load (4.5) and the comparison restated in per-flight units that do not depend on the fitted form.
3. **Every positive equilibrium of the delay-free reduction (M) is locally asymptotically stable, for all parameters** (Theorem 5.10) and numerically also in the delay model at six points of the band. Khoury et al. [22] give no stability analysis of the model with food.
4. **The two boundaries and two bounds are closed, ordered and interpretable.** They share the bookkeeping form (5.30) and come in a fixed order (Proposition 5.32). Global collapse for $p < p_E$ is proved for the reduced and for the published delay model (Theorems 5.17 and 5.20) and, with rationing at the empty comb, for every solution of

(M) (Proposition 5.19), where Khoury et al. [22] give plots and Ratti et al. [36] a simulated critical loss rate.

5. **In (M), the positive equilibrium is lost by a collision with the food axis, not at a fold.** The axis changes from transversally attracting to repelling at f_{crit} (Lemma 5.25, Theorem 5.26). The closed form of p^* comes from the same quadratic, so the branch meets the axis exactly there for every m (Propositions 5.6 and 5.27). A collapsing colony comes to rest with at most f_{crit} in the comb (Proposition 5.31).

Point 4 complements Betti et al. [4, p. 1588], who prove by a Laplace-transform method global stability of a linearised reduced colony model with sufficient food and conclude that a colony recovers if the cause of its losses is removed in time. The present work asks where recovery stops once food is limiting and answers with an exact budget identity and an explicit bound.

7.2 Comparison with the simulation literature

The calibration of Chapter 4 and the simulations of Requier et al. [37] rest on the same field campaign, so the two treatments can be compared directly. They fed both impacts into BEEHAVE, graded with the simulated load: the maximal foraging distance fell to zero at the highest loads and forager mortality rose from $1.00 \cdot 10^{-5}$ to at most $2.00 \cdot 10^{-5}$ per second of foraging, a doubling compared below with the increment derived here. Both changes act only between 28 August and 6 November. Of 993 simulations, 55.3% eventually reached the collapse endpoint.

This doubling is the one place in the literature where the mortality side of the same mechanism was given a number. The added 10^{-5} per second of flight equals the largest μ of this model, (4.5), after 36 seconds of flight per bee-day, so any realistic foraging time puts it well above the increment derived here: BEEHAVE assigns a mortality increase, this thesis derives one from the measured capture rate. Two of their findings bear on this thesis.

1. **Collapse happened after the pressure, not during it.** Only 2.5% collapsed while the hornets were present. The rest failed between 13 January and 1 May.
2. **At high loads the adult population rose.** Their path analysis splits the simulations at 13.3 hornets. Above that, hornet load has a *positive* effect on adult population, larvae and pollen storage but a negative effect on honey reserves.

O'Shea-Wheller et al. [33] add an observation from the other side: in *Bombus terrestris* colonies, where the hornets' predation attempts all failed, colony weight still fell with local hornet density, without substantial foraging inhibition at the nest and by a mechanism the authors leave open. It is a bumblebee, so it is quoted for the direction only.

Point 1 is compatible with this thesis: the compartment model, which has no winter, says that a colony ends the season lighter above 5.5 to 6.8 hornets at $\theta = 0$ (Table 6.3). BEEHAVE, which models the winter, sees such colonies fail months later.

Point 2 is the disagreement, in the *sign* of the population response at high load. It comes from one modelling decision, the share θ of baseline mortality that scales with flight activity. Section 6.3 gives the variant that reproduces their sign and its cost in hornets.

Remark 7.1 (A measurement that would decide it). Count adults in an apiary under sustained heavy pressure, through the autumn. If the adult population rises while the stores fall, (6.6) is the better description of the mortality side and the thresholds of this thesis are too low. If it falls, the model as written is. A difference in sign is the easiest kind to test.

7.3 Limitations and open questions

1. **No seasonality.** V is held constant. A seasonal $V(t)$ would make the thresholds properties of an annual map.
2. **One store.** Pollen and nectar share one variable, so a shortage of either looks the same.
3. **The mortality split θ** is unknown. All thresholds in hornets and the season loads are its pessimistic end (Section 6.3).
4. **Calibration.** One metric for V , one field study, seven stated assumptions (Table 4.1), among them a plentiful-forage intake c . Nothing beyond twenty hornets is data.
5. **The reduction.** Equilibria, the budget identity and collapse carry over to the delay model, local stability only numerically (Table 5.1). The food-axis results are for (M).
6. **Empty combs.** The rationing rule of Proposition 5.19 is a closure, not an observation.
7. **The window $V^* < V < V_E$ and the basins.** For solutions with convergent stores, collapse there is equivalent to eventually low stores (Propositions 5.33 and 5.31), which is not proved. The basin boundary is computed only numerically (Figure 5.5).

7.4 Outlook

Seasonality. A seasonal $V(t)$ turns (M) into a forced system with the present results as its autonomous limit. The first question is whether the slow relaxation of Table 5.1 survives forcing.

Closing the window. Proposition 5.33 reduces the window to one step: show that for $p < p^*$ a solution cannot keep its stores above f_{crit} at arbitrarily late times, perhaps with a food-dependent weight (Remark 5.28).

Other behavioural stressors. Any stressor acting on foraging activity occupies the same axis of the (p, m) plane. Whether behavioural and lethal stressors add up there is open.

7.5 Conclusion

Q1. Can foraging paralysis alone, without any increase in mortality, drive a colony to collapse at hornet loads inside the observed range? Yes, in the model, under sustained pressure and for $\theta \lesssim 0.95$ (Section 6.3). With m held at m_0 the positive equilibrium ceases to exist for $p < p^*$, that is above 8.07 hornets (Propositions 5.5 and 5.6). For $p < p_E$, above 9.73 hornets, every solution that stays in the model's domain converges to the food axis (Theorem 5.17). If an empty comb forces the colony to live on its intake, every solution does (Proposition 5.19). Between the two, collapse is proved while the stores stay low (Proposition 5.33), but in general it is open. None of these statements uses the mortality channel. *Sustained* means indefinitely: near V_E extinction takes longer than a season.

Q2. Is there a threshold and can it be stated explicitly and interpretably? Yes: two boundaries and two bounds, each in closed form. The break-even condition says a forager must bring in her own upkeep and the honey her replacement eats as a larva. It is necessary for persistence and free of the laying rate, the recruitment parameters and the half-saturation constants. The other three charge her with successively more hive bees in the same bookkeeping and they come in a fixed order (Proposition 5.32). The gap between the collapse bound and the existence boundary is the single factor $s^* = 0.28$ (5.31).

Q3. Which of the two channels can reach the existence boundary at field-realistic loads? Only the behavioural one. Predation moves the boundary by a hundredth of a hornet in

the fitted relation and by at most a tenth at the capture rates recorded on video. Captures are rare even at their fitted maximum and the highest risk per flight comes only with thin traffic. The first accounts for a factor of about 50, the second for about 17.

In short: from about six hornets (numerically 5.5 to 6.8) a colony loses honey over the season, above eight it has no positive equilibrium and above 9.73 extinction under sustained pressure is proved. These loads are pessimistic for the mortality split, rising to about 11, 17 and 21 hornets (Table 6.2), for forage they are optimistic (Remark 6.1). None of this depends on hornets killing bees.

Appendix A

Parameters and their sources

This appendix gives the source of every parameter value, first for the colony and then for the hornet.

A.1 Provenance of the colony parameters

Table A.1: Every value entering Chapters 3 to 6 that originates in Chapter 3, with its source and the step that produced it. Entries marked *derived* are computed in this thesis from the values above them.

quantity	value	source and derivation
L	2000 d ⁻¹	laying rate, Cramp [9] via Khoury et al. [21]
$\alpha_{\min}$	0.25 d ⁻¹	foraging as early as day four [13], so 1/4 [21]
$\alpha_{\max}$	0.25 d ⁻¹	doubling at zero food chosen by Khoury et al. [22], direction from Schulz et al. [46]
σ	0.75 d ⁻¹	$\sigma = 3\alpha_{\min}$, reversion above forager fraction 1/3, an assumption of Khoury et al. [21], reversion observed by Robinson et al. [38]
w	27 000	Khoury et al. [21, Fig. 2], no external source
v	5000	half of eggs fail at 5000 nurses [22]
b	500 g	low-store effects gone above 1 kg, $s(b) = \frac{1}{2}$
φ	1/9 d ⁻¹	nine days from laying to pupation
τ	12 d	capped pupal stage
c	0.10 g d ⁻¹	June maximum of Russell et al. [44] from Harbo [14], plentiful forage, Tab. 4.1(g)
γ_A	0.007 g d ⁻¹	Harbo [15]
γ_B	0.018 g d ⁻¹	0.163 g per worker reared to the pupal stage [15] over nine uncapped days: 0.163/9
m_0	0.154 d ⁻¹	reciprocal of the 6.5-day flightspan of colony 2, Khoury et al. [21, Tab. 1] after Rueppell et al. [43]
κ	0.1620 g	<i>derived</i> , γ_B/φ (Section 5.3)
k	23.1429	<i>derived</i> , $\gamma_B/(\gamma_A\varphi)$
m_c (2011)	0.355 d ⁻¹	critical death rate of the 2011 model, Khoury et al. [21]
$Q(m_0)$	2.8334	<i>derived</i> , positive root of (C.1)
$P(m_0, c)$	9.7217	<i>derived</i> , Eq. (5.2) at $p = 1$
$P(m=0.42)$	3.5657	<i>derived</i> , Eq. (5.2) at $p = 1$
f^*, H^*, F^*	1303.7 g, 9802.4, 2749.1	<i>derived</i> , Sec. C.1.2, consistent with Khoury et al. [22, Fig. 2c]
boundary in m	0.4010 d ⁻¹	<i>derived</i> , Prop. 5.4
m^*	0.4624 d ⁻¹	<i>derived</i> , $H^* = 0$ in (5.4)
(A8)	0.5741 d ⁻¹	Khoury et al. [22], = 31/54, $p_\kappa(m) = 1$: sufficient for extinction, not the boundary

Four values that need more than a table line.

- $w = 27\,000$ bees, the half-saturation constant of the 2011 model. No source is given. Khoury et al. [21, Fig. 2] plot E for $w = 4000, 10\,000$ and $27\,000$. They use the largest.

- $v = 5000$ bees. Khoury et al. [22] report this as congruent with their own experimental colonies and above the minimum population at which colonies rear brood effectively. Rosenkranz [42] name 5 000 bees as the critical strength for overwintering. The survival threshold of Becher et al. [2] rises to a comparable size at Khoury’s laying rate. The value is therefore at that threshold rather than comfortably above it. The two models count the population differently, so the comparison is indicative.
- $\gamma_B = 0.018$ g per brood item per day. Only larvae feed, but eggs and larvae share one compartment, so the average runs over the whole stage.
- $m_0 = 0.154 \text{ d}^{-1}$. The reciprocal of the 6.5-day flightspan of colony 2 in Khoury et al. [21, Tab. 1], measured by Rueppell et al. [43]. Table A.2 gives what the other three colonies of that table would produce.

Remark A.1 (The published parameter set is not unique). The values above are those of Khoury et al. [22, Figs. 2 and 3] and are used throughout. Their Table 1 fits the colony data of Harbo [14] with a different set ($\sigma = 1.3$, $c = 0.09$, $m = 0.06$, $\tau = 21$), whose numbers are not comparable with the ones computed here. None are used.

Table A.2: The four colonies of Khoury et al. [21, Tab. 1] with flightspans measured by Rueppell et al. [43] and the thresholds computed from each. The behavioural thresholds move by 1.5 to 2.1 hornets across the range. The last column, the distance to the existence boundary on the mortality axis over the largest predation increment (4.5), falls as m_0 rises: short-lived foragers are the least favourable case for both comparisons.

colony	flightspan [d]	$m \text{ [d}^{-1}\text{]}$	V_{acc}	V^*	V_E	$f_{\text{crit}} \text{ [g]}$	$(m^* - m_0)/\max \mu$
4 (Small)	8.8	0.114	7.56	9.96	11.86	281.6	961
1 (Large)	7.5	0.133	6.80	9.01	10.78	297.4	909
3 (Small)	6.7	0.149	6.21	8.28	9.97	309.5	865
2 (Large)	6.5	0.154	6.04	8.07	9.73	313.1	851

A.2 Provenance of the hornet parameters

The following items from Chapter 4 are carried here in full. The table at the end of this section lists every value of that chapter with its source.

The survey behind V . Table A.5 gives the size of the survey, the range of loads and the seasonal window. Three details of the protocol do not fit into it. The observation lasted 17 minutes, of which the first 15 gave the hornet count and the captures. The last two counted returning bees and gave the flight activity. One observer did all 603 observations, from July to December, to avoid an observer effect.

The published correction to Perry et al. [34]. The correction [35] fixes sign errors in the age-dependent functions $N(a)$ and $T(a)$ and a coefficient in $M(a)$. It leaves c_T and the sentence quoted in Section 4.3.4 untouched. Myerscough and Barron are coauthors of both Khoury et al. [22] and Perry et al. [34], so the two calibrations use one convention.

The two statistical models. The flight-activity curve of Requier et al. [37, Fig. 2a] is a generalized linear mixed model with Poisson error structure, computed with `glmer`, with region nested within year as random effects. It is highly significant ($Z = -20.65$, $P < 0.001$). The homing failure of Requier et al. [37, Fig. 2b] is modelled by a binomial GLMM with logit link, hour nested within date. It depends significantly and negatively on flight activity ($Z = -5.37$, $P < 0.001$).

The video study behind HF. A stereovision camera sat 50 cm above the flight board of one colony near La Rochelle and recorded from 16 to 25 October 2015, nine hours a day. That is 90 h, summarised in 255 slots of 15 minutes. Of 603 259 extracted flight trajectories, 5181 were classified as hornet by clustering on flight dynamics and body size. Every slot holding both species was reviewed by hand, twice, with unclear events discarded. What remained was 126 confirmed captures.

The capture count as a check on μ . The video study gives absolute numbers. 126 captures in ten days of nine hours are 12.6 captures a day. The model predicts μF captures a day, so with the bound (4.5) matching 12.6 a day would take at least 35 000 foragers. The size of the video colony is not available here. For a forager population between 3 000 and 11 000, the range of the model itself, the observed count corresponds to $\mu_{\text{obs}} = 12.6/F$ between $1.1 \cdot 10^{-3}$ and $4.2 \cdot 10^{-3} \text{ d}^{-1}$, that is 0.7 to 2.7 % of m_0 and three to twelve times the bound.

The traffic shows the same tension. If about half of the 598 078 bee trajectories (all trajectories less the hornets) were incoming flights, the mean homing failure was $126/299\,039 = 4.2 \cdot 10^{-4}$, which the fitted curve reaches at $p = 0.26$. At that activity a forager makes $\nu p = 0.77$ trips a day and about 30 000 incoming flights a day would then imply some 39 000 foragers. Both numbers point the same way: at that colony the activity scale of Fig. 2b, or the trip rate, differs from the survey's, which is assumption (f) of Table 4.1. Neither changes the comparison of Section 6.2. At μ_{obs} the predation channel is still short of $m^* - m_0$ by a factor of 73 to 270. That comparison is the counterfactual of step 3 in Section 6.2, predation alone at full traffic.

Table A.3: The two channels at the thresholds of Chapter 5, computed from (4.1) and (4.4) with $\beta = 0.109$ and $\nu = 3$. Channel 2 moves p across most of its range. Channel 1 stays below a quarter of a per cent of m_0 everywhere and turns around at $V = 16.6$.

V	$p(V)$	HF(p)	νp	$\mu [\text{d}^{-1}]$	μ/m_0
6.00	0.520	$8.43 \cdot 10^{-5}$	1.56	$1.32 \cdot 10^{-4}$	0.085 %
8.07	0.415	$1.60 \cdot 10^{-4}$	1.24	$1.99 \cdot 10^{-4}$	0.129 %
12.60	0.253	$4.28 \cdot 10^{-4}$	0.76	$3.25 \cdot 10^{-4}$	0.211 %
16.60	0.164	$7.37 \cdot 10^{-4}$	0.49	$3.62 \cdot 10^{-4}$	0.235 %
20.00	0.113	$1.01 \cdot 10^{-3}$	0.34	$3.41 \cdot 10^{-4}$	0.221 %

Remark A.2 (A reading of Requier et al. [37] that has to be corrected). Requier et al. [37] write that complete paralysis is “theoretically expected for at least 12.6 hornets simultaneously hovering in front of the beehives” and in the same sentence refer to the prediction 95 % confidence interval of their Fig. 2a. The number 12.6 is therefore where the *lower* confidence bound reaches zero, which requires an interval on the response scale. On the link scale its lower end would be a positive exponential too. It is not a zero of the fitted curve.

The fitted curve has no zero. A log link gives a strictly positive exponential. At $V = 12.6$ it still predicts

$$p(12.6) = e^{-0.109 \cdot 12.6} = 0.253,$$

that is 25 % residual activity and 22 % to 26 % across the β interval. Reading 12.6 as the zero of the response forces a linear $p(V)$ with intercept $V_c = 12.6$. That curve is too high at small V and too low at large V and it displaces every threshold derived from it. This thesis uses the exponential form.

Reported trip rates. Table A.4 collects the published values of ν and the bound (4.5) each of them implies.

Table A.4: Reported numbers of foraging trips per day. RFID studies usually average over bees that flew that day, Perry et al. [34] over all bees. The value $\nu = 3$ lies inside the measured range. The bound is linear in ν , so the extreme row covers every smaller value. Last column: the upper bound (4.5) on the mortality channel as a percentage of m_0 . The conclusion of Chapter 6 holds in every row.

ν [d ⁻¹]	source and method	definition	max μ/m_0
3	Perry et al. [34], RFID	all bees, all days	0.24 %
2.91 ± 2.50	Rodney and Kramer [39], five studies, $n = 2713$	per flying bee	0.23 %
≈ 4	Klein et al. [23], 17–21 trips in 4.2–4.9 d	per career day	0.31 %
4.6 ± 0.2	Minahan and Brunet [27], RFID	per flying bee	0.36 %
≈ 7	Visscher and Dukas [52], deaths per day over per trip	per foraging day	0.55 %
24	highest natural value for one bee-day [40]	maximum	1.88 %

Table A.5: Every value introduced in Chapter 4, with its source and the step that produced it.

quantity	value	source and derivation
V	0–20	max. hornets hovering simultaneously in 15 min at 3–5 m, 603 observations, 131 colonies, 75 apiaries, 2012–2016 [37]
window	28 Aug – 6 Nov	loads above five hornets occur only here [37]
β	0.109 hornet ⁻¹	<i>derived</i> , least squares through the origin on Tab. B.1, digitised from Requier et al. [37, Fig. 2a], Eq. (B.1) gives 0.1098
β spread	[0.107, 0.120]	<i>derived</i> , digitisation spread of the pointwise estimates, not a statistical interval
$p(12.6)$	0.253	<i>derived</i> , Rem. A.2. The 12.6 of Requier et al. [37] is the lower 95 % bound, not the curve manually confirmed in 90 h of video, 255 slots [37]
126 captures	—	
$\text{HF}_{\max}$	0.20 %	<i>derived</i> , Tab. B.2 from Requier et al. [37, Fig. 2b], fit gives 0.200 %
ζ (exponent)	6.09	inside the pointwise interval [6.01, 6.14] of the same digitisation. Least-squares fit 6.03 (Tab. B.2), effect ≤ 1 % on (4.5), below 0.001 hornets on every threshold
ν	3 d ⁻¹	mean trips per day over <i>all</i> bees including those that did not fly, RFID, $n = 118$ bees, from Perry et al. [34, p. 3432], who set $c_T = 0.033$ g per trip in the same model family
c_T	33.0 mg	load per trip [34], obtained there as 0.1 g over three trips, the decomposition (4.3)
trips per career	17–21	Klein et al. [23], whole-career tracking in two colonies, 17 ± 1 and 21 ± 1 , over foraging spans of 4.2 and 4.9 d: about four trips a day (Tab. A.4)
$\mu(8.07)/m_0$	0.129 %	<i>derived</i> , Tab. A.3
$\max_V \mu$	$3.62 \cdot 10^{-4}$ d ⁻¹	<i>derived</i> , Eq. (4.5), = 0.235 % of m_0 , attained at $V = 16.6$
required/attainable	≈ 851	<i>derived</i> , Eq. (6.4)
p^* with μ	0.4155	<i>derived</i> , against 0.4151 at $\mu \equiv 0$, a shift of 0.01 hornets
p^* numerical	0.3879–0.4104	<i>derived</i> , Tab. B.3, horizon-dependent, converging to 0.4151

Appendix B

Numerical protocol

All computations were done in Python: ordinary differential equations with LSODA from SciPy, the delay model with a fixed-step fourth-order Runge–Kutta scheme on a stored history, closed forms by bisection on the cascade of Lemma 5.3 and the algebraic identities of Chapter 5, including the reduced trace (5.13) and the closed form (5.10), against numerically evaluated Jacobians and roots. Every numerical statement in the text is a control and not a proof. It carries its integrator, tolerance and horizon where it is stated. The code that produces every table and figure is available from the author on request.

B.1 Digitisation of the calibration figures

The least-squares estimate for a line through the origin in the coordinates $(V, -\ln p)$, on the values of Table B.1, is

$$\begin{aligned}\beta &= \frac{\sum_i V_i (-\ln p_i)}{\sum_i V_i^2} = \frac{5(0.5978) + 10(1.1087) + 15(1.6094) + 20(2.2073)}{25 + 100 + 225 + 400} \\ &= \frac{82.363}{750} = 0.1098.\end{aligned}\tag{B.1}$$

Table B.1: Digitisation of Requier et al. [37, Fig. 2a] and the resulting estimates of β . The pointwise estimates scatter by about 12 % around the least-squares value. That scatter measures the digitisation, not the statistical uncertainty of the fitted curve.

V	p read off	$-\ln p$	$\beta_i = -\ln p_i/V_i$	fit $e^{-0.1098V}$
0	1.00	0	—	1.000
5	0.55	0.5978	0.1196	0.577
10	0.33	1.1087	0.1109	0.334
15	0.20	1.6094	0.1073	0.193
20	0.11	2.2073	0.1104	0.111

Table B.2: Digitisation of Requier et al. [37, Fig. 2b]. The abscissa is flight activity in per cent, so $p = x/100$. The ordinate is homing failure in per cent. The pointwise estimates $\zeta_i = -\ln(\text{HF}_i/\text{HF}_0)/p_i$ lie between 6.01 and 6.14.

x [%]	p	HF read off [%]	ζ_i	fit $0.200 e^{-6.03p}$ [%]
0	0.0	0.202	—	0.200
10	0.1	0.110	6.078	0.110
20	0.2	0.060	6.070	0.060
30	0.3	0.032	6.142	0.033
40	0.4	0.018	6.045	0.018
50	0.5	0.010	6.011	0.010

A least-squares fit of $\ln \text{HF}$ on p over Table B.2 gives

$$\text{HF}(p) = 0.00200 e^{-6.03p},$$

against the values 0.0020 and 6.09 used in (4.1). The prefactor agrees to 0.2 % and the exponent to 1.1 %. The value 6.09 lies inside the pointwise interval $[6.01, 6.14]$. This thesis keeps 0.0020 and 6.09. With 6.03 the bound (4.5) rises by one per cent.

B.2 The delay reduction, numerically

Protocol. Each model is integrated with fixed-step RK4 at $\Delta t = 0.02$ d from the published initial state $(B, H, F, f) = (0, 16\,000, 8\,000, 0)$, at $m = m_0$, to a horizon T . The three-dimensional model starts with $f_0 = 800$ g, which moves its threshold by 0.09 % against $f_0 = 0$. A colony counts as persisting if $H + F > 200$ at $t = T$. The threshold is located by 20 bisection steps on $p \in [0.30, 0.70]$, which brackets it to $4 \cdot 10^{-7}$. The delay uses the stored history of B on the integration grid. For $\tau = 0$ the model is integrated as an ordinary differential equation, with $B(t - \tau)$ replaced by $B(t)$.

Table B.3: Threshold in p below which the colony fails to persist, under the protocol above. Deviations are relative to the delay model at the same horizon. The step size does not matter (the delay model gives 0.4031 at $\Delta t = 0.04, 0.02$ and 0.01 d), but the horizon does.

horizon T	4D delay, $\tau = 12$	4D ODE, $\tau = 0$, B a state	3D model (M)
1000 d	0.3879	0.3981 (+2.6 %)	0.3983 (+2.7 %)
2000 d	0.4031	0.4084 (+1.3 %)	0.4086 (+1.4 %)
4000 d	0.4104	0.4129 (+0.6 %)	0.4130 (+0.6 %)
analytic (Sec. 5.1)	$p^* = 0.4151$		

The detected threshold rises with the horizon and approaches p^* from below, as it must: just below p^* the decay towards extinction is arbitrarily slow, so every finite-horizon criterion underestimates p^* . The deviation of the reduction falls from 2.7 % to 0.6 % in step with the delay model’s own convergence, as Remark 4.2 requires. Keeping B as a state with $\tau = 0$ tracks the reduction to within 0.02 percentage points.

Linear stability of the delay model. Linearising (K), with the substitutions of Section 4.1, at the common equilibrium gives $\dot{y} = A_0 y + A_1 y(t - \tau)$ for $y = (B, H, F, f)$, where A_1 has the single entry φ in the H -row and B -column, with characteristic function $D(\lambda) = \det(\lambda I - A_0 - A_1 e^{-\lambda \tau})$. A root with $\operatorname{Re} \lambda \geq 0$ satisfies $|\lambda| \leq \|A_0\| + \|A_1\|$, which is below 3 at every tested point. The roots in $\{\operatorname{Re} \lambda \geq \sigma_0, |\lambda - \sigma_0| \leq 30\}$ were counted by the argument principle on its boundary ($1.6 \cdot 10^5$ points, dense near the real axis). At each of the six points of Table 5.1 there is no root for $\sigma_0 = 0$ or for $\sigma_0 = \lambda_D/2$ and exactly one for $\sigma_0 = 1.5 \lambda_D$, where $\lambda_D < 0$ is the real root found by Newton’s method from the slow eigenvalue of (M), so λ_D is the rightmost root. As a control, the nonlinear delay model integrated from a perturbed equilibrium (RK4, $\Delta t = 0.02$ d, 3000 d) decays at the rate λ_D to six significant figures at $p = 0.45$ and 0.48 .

B.3 The boundaries of the (p, m) plane

Table B.4: The three boundaries of the plane, from Lemma 5.3 and Proposition 5.5 solved at fixed m (derived, not a new result). The band is narrow and almost of constant width in p and it slides to the right as mortality grows. Hornet loads with $\beta = 0.109$, given where $p \leq 1$.

m [d ⁻¹]	$p^*(m)$	$p_{\text{acc}}(m)$	width	$p_E(m)$	V^*	V_{acc}
0.1540 = m_0	0.4151	0.5178	0.1027	0.3464	8.07	6.04
0.2000	0.5036	0.6083	0.1047	0.4290	6.29	4.56
0.2500	0.5990	0.7062	0.1072	0.5187	4.70	3.19
0.3000	0.6939	0.8037	0.1098	0.6085	3.35	2.00
0.3500	0.7884	0.9010	0.1126	0.6982	2.18	0.96
0.4010	0.8846	1.0000	0.1154	0.7898	1.13	0.00
0.4624 = m^*	1.0000	1.1188	0.1188	0.9000	0.00	—

B.4 The food axis, residual food and the open window

Table B.5 checks the transverse growth rate of Lemma 5.25 in the full system. Table B.6 gives the food left in the comb when a colony dies from the healthy state, the numbers behind Section 5.3.4. Every value lies below f_{crit} , as Proposition 5.31 requires.

Table B.5: The transverse growth rate $\lambda(f)$ of Lemma 5.25 against the exponential rate measured in the full nonlinear system (M), at $m = m_0$. Initial population $N_0 = 10^{-4}$, horizon 20 d, LSODA with $\text{rtol} = 10^{-12}$. Numerical control.

f [g]	λ analytic	measured	difference
250	-0.027012	-0.027012	$9.9 \cdot 10^{-9}$
313	-0.000052	-0.000052	$9.5 \cdot 10^{-9}$
400	+0.038377	+0.038377	$6.2 \cdot 10^{-9}$
600	+0.115474	+0.115474	$1.2 \cdot 10^{-8}$
800	+0.168960	+0.168960	$4.1 \cdot 10^{-8}$

Table B.6: Residual food after collapse at $m = m_0$, from (9000, 2700, 1300), $T = 6000$ d, LSODA, $\text{rtol} = 10^{-10}$. Numerical observation, not a consequence of Theorem 5.17. At $p = 0.100$ the solution leaves Ω after 16 d. With rationing it collapses with $f_\infty = 0$ (Proposition 5.19).

p	V	f_∞ [g]	of f_{crit}
0.100	21.12	leaves Ω	—
0.1157	19.79	0	—
0.150	17.40	104.5	33.4 %
0.200	14.77	172.1	55.0 %
0.250	12.72	216.0	69.0 %
0.300	11.05	250.6	80.0 %
0.319	10.48	262.2	83.8 %

B.4.1 The window between V^* and V_E

Section 5.5 states that no positive equilibrium exists above $V^* = 8.07$ and that collapse is proved only above $V_E = 9.73$. In between, the claim that colonies die is numerical. Table B.7 documents it, for four initial states (H_0, F_0, f_0) spanning two orders of magnitude in population: A = (20 000, 8 000, 5 000), B = (9 000, 2 700, 1 300), C = (2 000, 600, 600) and D = (200, 80, 400).

Table B.7: Time until $H + F < 1$ inside the window from states A to D, with $m = m_0 + \mu(V)$. LSODA, $\text{rtol} = 10^{-10}$, $\text{atol} = 10^{-12}$, horizon $2 \cdot 10^5$ d. All twenty runs collapse. Numerical control, not a proof: it rules out survival from these four states, not a limit cycle. The times grow towards V^* (Remark 5.11). The residual food barely depends on the initial state and lies below f_{crit} (Proposition 5.31). The last row lies at the collapse bound: with μ included, $V_E = 9.71$ (Table 6.2).

V	p	time to $H + F < 1$ [d]				residual
		A	B	C	D	f [g]
8.10	0.4136	16 704	16 565	16 138	14 591	312.3
8.50	0.3959	2 207	2 103	1 902	1 580	303.8
9.00	0.3749	1 127	1 047	924	778	293.2
9.50	0.3550	770	704	616	536	282.6
9.72	0.3466	678	617	539	478	278.0

Appendix C

Reproduction of the Khoury model

This appendix reproduces the published results of Khoury et al. [22] and documents the two errors in their Appendix S1 and the critical death rate referred to in Chapter 3.

C.1 Reproduction of the published results

Appendix S1 of Khoury et al. [22] states the equilibria in two regimes and two critical parameter values. All of it is rederived here. By Remark 4.2 the equilibria of (K) are those of (M), so Lemma 5.3 at $p = 1$ does most of the work.

C.1.1 The food-abundant regime

If $f \rightarrow \infty$, then $s(f) \rightarrow 1$ and $b^2/(b^2 + f^2) \rightarrow 0$. System (K) then reduces to (A1)–(A3) of the appendix. The age ratio $Q = H/F$ is fixed by (5.5). Written in $Q = 1/J$ it reads

$$\alpha_{\min} Q^2 + (\alpha_{\min} - \sigma - m) Q - m = 0, \quad (\text{C.1})$$

which is (A5), with the single positive root $Q = 2.8334$ at $m = m_0$. The size follows from step (iii) of Lemma 5.3 with $s = 1$:

$$F = \frac{L}{m} - \frac{v}{Q} = \frac{LQ - mv}{mQ}, \quad H = QF, \quad B = \frac{m}{\varphi} F, \quad (\text{C.2})$$

which is (A4). The brood expression is Lemma 5.2.

C.1.2 The food-limited regime

Keep the food-dependent terms. The age ratio is P of (5.2) at $p = 1$. Step (ii), (5.3) with $J = 1/P$, fixes the food level. Write $X := b^2/(b^2 + f^2) \in (0, 1]$, so that $1/X = 1 + f^2/b^2$ and $f = b\sqrt{1/X - 1}$. Step (ii) gives $X = [m/P + \sigma/(1 + P) - \alpha_{\min}]/\alpha_{\max}$, hence

$$f = b \sqrt{\frac{\frac{\alpha_{\max}}{m}}{\frac{1}{P} + \frac{\sigma}{P+1} - \alpha_{\min}} - 1} \quad (\text{C.3})$$

and, from step (iii), (5.4),

$$F = \frac{Lf^2}{m(f^2 + b^2)} - \frac{v}{P}, \quad H = PF, \quad B = \frac{m}{\varphi} F \quad (\text{C.4})$$

Together these are (A6), with one difference in (C.3) that Section C.1.3 takes up. The expression requires $0 < m/P + \sigma/(1 + P) - \alpha_{\min} < \alpha_{\max}$. The lower bound keeps f finite, the upper keeps the radicand positive.

Numerical evaluation at the reference point. At $m = 0.42$ and $c = 0.1$, the parameters of Khoury et al. [22, Fig. 2c], the chain (5.2), (C.3), (C.4) runs as follows:

$$\begin{aligned}
P &= \frac{0.1}{0.007} - 1 - \frac{0.018 \cdot 0.42}{0.007 \cdot (1/9)} = 14.2857 - 1 - 9.7200 = 3.5657, \\
\frac{m}{P} + \frac{\sigma}{1+P} - \alpha_{\min} &= 0.11779 + 0.16427 - 0.25 = 0.03206, \\
X &= 0.03206/0.25 = 0.12824, \quad f = 500\sqrt{1/0.12824 - 1} = 1303.7 \text{ g}, \\
s(f) &= 0.87175, \quad H = \frac{2000 \cdot 0.87175}{0.42/3.5657} - 5000 = 9802.4, \\
F &= H/P = 2749.1, \quad B = \frac{m}{\varphi} F = 10\,391.6
\end{aligned}$$

They are consistent with the published figure, which shows curves but prints no numbers. Integrating (K) directly, with the delay in place, returns them to five significant figures.

C.1.3 Two errors in Appendix S1

(a) A sign in the denominator of (A6). The appendix prints the denominator of (C.3) as

$$\frac{\sigma}{P+1} - \frac{m}{P} - \alpha_{\min} \quad \text{instead of} \quad \frac{m}{P} + \frac{\sigma}{P+1} - \alpha_{\min},$$

a difference of exactly $2m/P$. The derivation above gives the second form. The term m/P enters as the equilibrium value of R , so after rearrangement it stands on the opposite side from $\alpha_{\min}$ and $\alpha_{\max}X$, with a plus sign.

The arithmetic settles it. At the parameters of Khoury et al. [22, Fig. 2c] the printed denominator equals $0.16427 - 0.11779 - 0.25 = -0.20352$. It is negative, so $\alpha_{\max}/(\cdot) - 1 < 0$ and the printed formula returns no real value for f . It fails at the very point the accompanying figure displays. The same sign appears again in the running text a page later, so it is not one bad display.

(b) The quadratic for the food-limitation boundary. The appendix states that food stops being limiting when the denominator of (C.3) vanishes and that this happens when m is a root of ¹

$$\left(\frac{\gamma_B}{\gamma_A \varphi}\right) m^2 - \left(\frac{\sigma \gamma_B}{\gamma_A \varphi} - \frac{c}{\gamma_A}\right) m + \alpha_{\min} + \alpha_{\max} + \sigma - \frac{\sigma c}{\gamma_A} = 0 \quad (\text{C.5})$$

Three independent reasons say this cannot be right.

First, $\alpha_{\max}$ cannot occur. The denominator contains only $\alpha_{\min}$, σ , m and P . The parameter $\alpha_{\max}$ multiplies $b^2/(b^2 + f^2)$ in (5.3) and drops out the moment that factor is set to zero.

Second, the degree structure is wrong. Substituting $P = A - km$ with $A := c/\gamma_A - 1$ and $k := \gamma_B/(\gamma_A \varphi)$ into $\alpha_{\min} P^2 + (\alpha_{\min} - \sigma - m)P - m = 0$ and expanding gives

$$k(\alpha_{\min} k + 1) m^2 - [2\alpha_{\min} A k + (\alpha_{\min} - \sigma)k + A + 1] m + A(\alpha_{\min} A + \alpha_{\min} - \sigma) = 0 \quad (\text{C.6})$$

The coefficient of m^2 is $k(\alpha_{\min} k + 1)$ and contains k^2 . The printed equation has k alone. At the reference parameters $k = 23.1429$ and $A = 13.2857$, so the correct coefficient is 157.04 against the printed 23.14.

Third, the root is out of range. Equation (C.6) has roots 0.4010 and 0.5952 and only the first is attained, since the second lies above the threshold of Section C.1.4. Equation (C.5) has

¹Appendix S1 also assigns the label (A6) to two different displays. Here (A6) is the equilibrium of the food-limited regime.

roots -0.5766 and 0.7093 . The positive one lies above the appendix's own (A8) and far above any death rate at which a colony still exists. The value 0.4010 is consistent with Khoury et al. [22, Figs. 2b and 2c], where the stores still grow at $m = 0.3$ and settle at $m = 0.42$. The value 0.7093 is not.

The condition is better stated without a quadratic at all.

The denominator of (C.3) is the expression of step (ii), (5.3), written with $J = 1/P$, since $\sigma/(P+1) = \sigma J/(1+J)$. By Proposition 5.4 it vanishes if and only if $P = Q$.

C.1.4 The critical death rate

The appendix closes with a threshold:

$$m = \frac{\varphi \gamma_A}{\gamma_B} \left(\frac{c}{\gamma_A} - 1 \right) = \frac{(1/9) \cdot 0.007}{0.018} \cdot 13.2857 = \frac{31}{54} = 0.5741, \quad (\text{C.7})$$

which is (A8) and the sentence beside it says the colony goes extinct there, “where P becomes negative”. The derivation is immediate from (5.2) at $p = 1$. Khoury et al. call this value m^* , which is not the m^* of this thesis. It is exactly the break-even condition $p_\kappa(m) = 1$ of Lemma 5.12 on the mortality axis, so extinction beyond it is correct, as a sufficient bound. It does not, however, mark the loss of the positive equilibrium.

$P = 0$ means $H/F \rightarrow \infty$. It does not mean the colony is empty. The positive equilibrium disappears earlier and by a different mechanism. Raising m from m_0 at $c = 0.1$, four conditions are crossed in this order.

Table C.1: The order in which the equilibrium conditions bind, at $c = 0.1$. The one that fails first is $H^* > 0$, not $P > 0$. Values obtained by bisection on the closed-form cascade to twelve digits, so they carry no integration error.

condition	source	meaning	m
$P = Q$	Prop. 5.4	stores stop growing, f^* finite	0.4010
$H^* = 0$	Eq. (5.4)	equilibrium meets the food axis	0.4624
denominator $= \alpha_{\max}$	Eq. (C.3)	$(f^*)^2 = 0$, no real food level	0.4888
$P = 0$	Eq. (C.7)	the appendix's stated threshold	0.5741

The binding condition is $H^* > 0$ in (5.4) and it fails at

$$m^* = 0.4624 \text{ d}^{-1}, \quad \text{with} \quad f^*(m^*) = 449.8 \text{ g} \quad (\text{C.8})$$

There H^* and $F^* = JH^*$ vanish together while the stores stay finite. Between m^* and (C.7) the quantity P is still positive and yet no positive equilibrium exists.

The check that settles it is inside the same paper. The steady-state diagram Khoury et al. [22, Fig. 3a] shows the stores diverging below about $m = 0.40$ and the populations reaching zero at about 0.46 , which are 0.4010 and $m^* = 0.4624$, not 0.5741 . The trajectories of their Figs. 2c and 2d, at $m = 0.42$ and 0.5 , agree, but alone they cannot settle it, since the model is bistable.

C.1.5 Numerical reproduction of the published figures

System (K) was integrated with a fixed-step fourth-order Runge–Kutta scheme at step 0.02 d , from the published initial state $(B, H, F, f) = (0, 16\,000, 8\,000, 0)$, with m running over $0.1, 0.3, 0.42$ and 0.5 . All four regimes of Khoury et al. [22, Fig. 2] are recovered: unbounded stores beside a stable population at $m = 0.1$ and at $m = 0.3$, the finite stores computed in Section C.1.2 at $m = 0.42$ and extinction with food left in the comb at $m = 0.5$. The last is the regime the paper was written for and the model produces it without any assumption about disease.

Appendix D

Technical proofs

This appendix contains the proofs that Chapters 4 and 5 state without proof.

D.1 Reduction of the trace

This is the proof of Lemma 5.9. It is pure algebra on the entries of the Jacobian and the statement is what matters in the main text.

Proof. Three substitutions, all at a positive equilibrium, so $H > 0$ and division by H is legitimate. First, $Ls g = LS = mF = mJH$ and dividing by H and using $g = H/(H + v)$ gives $LS = mJ(H + v)$, hence

$$Ls \frac{v}{(H + v)^2} = \frac{mJ(H + v)v}{(H + v)^2} = mJ \frac{v}{H + v} = mJ(1 - g)$$

Second, $R = mJ$ by step (ii) and $\sigma HF/(H + F)^2 = \sigma u(1 - u) = \sigma J/(1 + J)^2$, while $\sigma H^2/(H + F)^2 = \sigma/(1 + J)^2$. Third, $s'(f) = 2s\psi/f$, since $s\psi = f^2 b^2/(f^2 + b^2)^2$. Also $\kappa Lg s = \kappa LS = \kappa mJH$, so

$$\kappa Lg s' = \frac{\kappa Lg s}{s} \cdot \frac{2s\psi}{f} = \frac{2\kappa mJH\psi}{f}$$

Hence

$$\text{tr } \mathcal{J} = \left[mJ(1 - g) - mJ - \frac{\sigma J}{(1 + J)^2} \right] + \left[-\frac{\sigma}{(1 + J)^2} - m \right] + \left[-\frac{2\kappa mJH\psi}{f} \right],$$

and $\sigma J/(1 + J)^2 + \sigma/(1 + J)^2 = \sigma(1 + J)/(1 + J)^2 = \sigma/(1 + J)$. Negating gives (5.13). All four terms are positive because $m, J, H, f, \psi, \kappa, \sigma > 0$ and $0 < H/(H + v) < 1$. $\square$

D.2 Homogeneity of the extinction limit

This is the proof of Proposition 5.22.

Proof. Substituting $H = (1 - u)N$ and $F = uN$ gives $S = sH/(H + v) = s(1 - u)N/((1 - u)N + v)$, so $LS = A_N(1 - u)N$ exactly. Then $\dot{N} = \dot{H} + \dot{F} = LS - mF$ gives (5.25). For the shape, $\dot{u} = \dot{F}/N - u\dot{N}/N$ with $\dot{F}/N = (1 - u)R - mu$ and $R = \alpha(f) - \sigma u$, so

$$\dot{u} = (1 - u)R - mu - u[A_N(1 - u) - mu] = (1 - u)[R - uA_N - mu] = (1 - u)[\alpha - u(\sigma + A_N + m)] \quad \square$$

D.3 The transverse growth rate

This is the proof of Lemma 5.25.

Proof. The zero. The left side q of (5.8) satisfies $q(1) = \omega^2 + \omega(\sigma - \alpha_{\min}) + m(\omega - \alpha_{\min}) > 0$, because $\omega = L/v > \alpha_{\min}$ and $\sigma > \alpha_{\min}$. With $q(0) < 0$ its positive root s^* lies in $(0, 1)$, so f_{crit} is finite. Substituting (5.27) into (5.28),

$$\lambda = A - u^*(A + m) = A - \frac{\alpha(A + m)}{\sigma + A + m},$$

so $\lambda = 0$ is equivalent to $A(\sigma + A + m) = \alpha(A + m)$, which is (5.11). With $A = \omega s$ and $\alpha = \alpha_{\min} + \alpha_{\max} - \alpha_{\max}s$ it becomes (5.8), whose only positive root is s^* (Proposition 5.6), so the only zero of λ is f_{crit} .

Monotonicity. λ depends on f only through A and α , with

$$\frac{\partial \lambda}{\partial A} = 1 - \frac{\alpha \sigma}{(\sigma + A + m)^2}, \quad \frac{\partial \lambda}{\partial \alpha} = -\frac{A + m}{\sigma + A + m} < 0$$

So $\partial \lambda / \partial A > 0$ holds exactly when

$$\alpha \sigma < (\sigma + A + m)^2, \quad (\text{D.1})$$

and this always holds. The model assumes $\sigma > \alpha_{\min} + \alpha_{\max}$ (Section 3.1.2), so $\alpha \leq \alpha_{\min} + \alpha_{\max} < \sigma < \sigma + A + m$ and trivially $\sigma < \sigma + A + m$. The product of the two smaller numbers is smaller than $(\sigma + A + m)^2$. At the reference parameters the left side is at most 0.375 and the right side at least $(\sigma + m_0)^2 = 0.8172$. With (D.1) and since A is strictly increasing in f while α is strictly decreasing, λ is strictly increasing in f . $\square$

The sign statement of Lemma 5.25 is a reading rather than a further claim: by (5.25), λ is the growth rate of N once u has settled at u^* , which it does for every $u_0 \in [0, 1]$ in the limit $N \rightarrow 0$ at fixed f . Theorem 5.26 makes this rigorous.

D.4 Stability of the food axis

This is the proof of Theorem 5.26. It works in the coordinates (N, u, f) of Proposition 5.22 and never tracks the shape u exactly. It only needs one-sided bounds on u , because the growth rate of N is decreasing in u : decay needs u bounded below, growth needs u bounded above and each bound comes from a barrier on which $\dot{u}$ has a fixed sign.

Proof. Write $G(u, f, N) := \alpha(f) - u(\sigma + A_N(u, f) + m)$ for the bracket of (5.26), so that $\dot{u} = (1 - u)G$ and $I := [f_0 - \eta, f_0 + \eta]$ with $0 < \eta < f_0$. It suffices to prove each part for all sufficiently small η , since the conclusion for one η implies it for every larger one. Put $K := \max_I A + m$ and $C := cp + \gamma_A + \kappa L/v$. Four elementary facts are used.

(a) *The reduction error.* From (5.24) and $0 \leq s \leq 1$,

$$0 \leq A(f) - A_N(u, f) = \frac{L s(f) (1 - u) N}{v [(1 - u)N + v]} \leq c_1 N, \quad c_1 := \frac{L}{v^2} = 8 \cdot 10^{-5}, \quad (\text{D.2})$$

uniformly in $u \in [0, 1]$ and $f \geq 0$.

(b) *Monotonicity in the shape.* The denominator of A_N falls as u rises, so uA_N is increasing in u and G is strictly decreasing in u .

(c) *The growth rate against the shape.* By (5.27) and the definition of λ , $A(1 - u) - mu = \lambda(f) - (A + m)(u - u^*(f))$. With (5.25) and (a) this gives the two-sided bound

$$\lambda(f) - (A + m)(u - u^*(f)) - c_1 N \leq \frac{\dot{N}}{N} \leq \lambda(f) - (A + m)(u - u^*(f)) \quad (\text{D.3})$$

(d) *The store drift.* $cpF \leq cpN$, $\gamma_A(H + F) = \gamma_A N$ and $\kappa LS \leq \kappa LH/v \leq \kappa LN/v$, so $|f| \leq CN$ on Ω and $|f(t) - f(0)| \leq C \int_0^t N$.

Finally one elementary fact, called the *barrier principle* below: if a differentiable function x has $x(0) \leq X$ and $\dot{x} < 0$ whenever $x = X$, then $x(t) \leq X$ for all $t \geq 0$, because x cannot cross the level X upwards at a point where it is decreasing.

(i) Let $f_0 < f_{\text{crit}}$. Since λ is continuous and increasing and u^* continuous, η can be taken so small that $\lambda \leq -\delta < 0$ on I and $K(\max_I u^* - \min_I u^*) \leq \delta/4$. Choose $\varepsilon_0 > 0$ with $K\varepsilon_0 \leq \delta/4$ and put $\underline{u} := \min_I u^* - \varepsilon_0$.

Step 1: the shape is soon bounded below. As long as $f \in I$ and $u \leq \underline{u}$, (b) and $A_N \leq A$ give

$$G(u) \geq G(\underline{u}) \geq \alpha - \underline{u}(\sigma + A + m) = (\sigma + A + m)(u^*(f) - \underline{u}) \geq (\sigma + m)\varepsilon_0,$$

so $\dot{u} \geq r := (1 - \underline{u})(\sigma + m)\varepsilon_0 > 0$. Hence $u \geq \underline{u}$ from the time $T_0 := 1/r$ on at the latest, whatever $u(0) \in [0, 1]$. By the barrier principle it stays there. No smallness of N is used.

Step 2: decay. For $t \geq T_0$ and $f \in I$ the upper bound in (D.3) gives $\dot{N}/N \leq -\delta + K(u^*(f) - \underline{u}) \leq -\delta + \delta/4 + \delta/4 = -\delta/2$. On $[0, T_0]$ it gives $\dot{N}/N \leq \max_I A =: A_I$, so $N(t) \leq N(0)M$ with $M := e^{A_I T_0}$.

Step 3: the stores stay close. As long as $f \in I$, Steps 1 and 2 give $\int_0^t N \leq N(0)M(T_0 + 2/\delta)$ and by (d) $|f(t) - f(0)| \leq CN(0)M(T_0 + 2/\delta)$. Choose $\rho < \min\{\eta/2, \eta/M, \eta\delta/[2CM(\delta T_0 + 2)]\}$. Then $|f(t) - f_0| < \eta$ for all t , so f never leaves I and the estimates hold for all t : $N(t) < \eta$, $N \rightarrow 0$ exponentially, $\dot{f}$ is integrable and $f \rightarrow f_\infty$ with $|f_\infty - f_0| < \eta$.

(ii) Let $f_0 > f_{\text{crit}}$ and $\lambda_0 := \lambda(f_0) > 0$. Take η so small that $\lambda \geq \lambda_0/2$ on I and $K(\max_I u^* - \min_I u^*) \leq \lambda_0/16$, choose $\varepsilon_0 > 0$ with $K\varepsilon_0 \leq \lambda_0/16$ and $\bar{u} := \max_I u^* + \varepsilon_0 < 1$, possible since $u^* \leq (\alpha_{\min} + \alpha_{\max})/(\sigma + m) < 1$. Finally choose $\rho_1 > 0$ with $c_1\rho_1 \leq \lambda_0/16$, $c_1\rho_1 \leq (\sigma + m)\varepsilon_0/2$ and $\rho_1 \leq \eta\lambda_0/(8C)$.

The shape stays bounded above. As long as $f \in I$, $N \leq \rho_1$ and $u \geq \bar{u}$, (b) and (a) give

$$G(u) \leq G(\bar{u}) \leq (\sigma + A + m)(u^*(f) - \bar{u}) + \bar{u}c_1N \leq -\frac{(\sigma + m)\varepsilon_0}{2} \quad (\text{D.4})$$

So $\dot{u} < 0$ on $\{u = \bar{u}\}$ and by the barrier principle a solution with $u(0) \leq \bar{u}$ keeps $u \leq \bar{u}$ as long as $f \in I$ and $N \leq \rho_1$.

Growth. In that phase the lower bound in (D.3) gives

$$\frac{\dot{N}}{N} \geq \frac{\lambda_0}{2} - K(\bar{u} - u^*(f)) - c_1\rho_1 \geq \frac{\lambda_0}{2} - \frac{3\lambda_0}{16} \geq \frac{\lambda_0}{4} \quad (\text{D.5})$$

The stores stay close. Start at $N(0) = \rho \in (0, \rho_1)$, $u(0) = u^*(f_0)$, $f(0) = f_0$, that is at $(H, F) = ((1 - u^*)\rho, u^*\rho)$, arbitrarily close to $(0, 0, f_0)$. While N grows at rate at least $\lambda_0/4$, $N(s) \leq N(t)e^{-\lambda_0(t-s)/4}$ for $s \leq t$, so $\int_0^t N \leq 4N(t)/\lambda_0 \leq 4\rho_1/\lambda_0$ and, by (d), $|f(t) - f_0| \leq 4C\rho_1/\lambda_0 \leq \eta/2$. Hence f stays in I , u stays below $\bar{u}$ and the phase can end only by N reaching ρ_1 , which happens before $t = 4\ln(\rho_1/\rho)/\lambda_0$. Since ρ is arbitrary and ρ_1 does not depend on it, $(0, 0, f_0)$ is unstable. $\square$

D.5 Limit point of a collapsing colony

This is the proof of Proposition 5.31. It reuses the barrier (D.4) and the growth bound (D.5) of Appendix D.4.

Proof. Since $\{H = 0\}$ is invariant and solutions are unique, $H(0) > 0$ gives $H(t) > 0$, so $u(t) < 1$, for all t . Suppose $\lambda(f_\infty) > 0$ and choose η , ε_0 , $\bar{u}$ and ρ_1 as in part (ii) of Appendix D.4, with $f_0 := f_\infty$ and $\lambda_0 := \lambda(f_\infty)$. Since $f \rightarrow f_\infty$ and $N \rightarrow 0$, there is T with $f(t) \in I$ and $N(t) \leq \rho_1$ for all $t \geq T$.

The shape falls below $\bar{u}$. While $u \geq \bar{u}$, (D.4) gives for $w := 1 - u$ the inequality $\dot{w} = -(1 - u)G \geq rw$ with $r := (\sigma + m)\varepsilon_0/2$. Since $w(T) > 0$, w grows at least like $e^{r(t-T)}$, so u reaches $\bar{u}$ in finite time. By the barrier principle it stays below $\bar{u}$ afterwards.

The population would grow. From then on (D.5) gives $\dot{N}/N \geq \lambda(f_\infty)/4 > 0$, so $N \rightarrow \infty$, which contradicts $N \rightarrow 0$.

Therefore $\lambda(f_\infty) \leq 0$ and since λ is strictly increasing with its only zero at f_{crit} (Lemma 5.25), $f_\infty \leq f_{\text{crit}}$. $\square$

D.6 The existence boundary moves with mortality

By (5.10), $p^*(m)$ is strictly increasing in m once $m/s^*(m)$ is. Write $q(s, m)$ for the left side of (5.8), $a := \omega^2 + \omega\alpha_{\max}$, $b_0 := \omega(\sigma - \alpha_{\min} - \alpha_{\max}) > 0$, $b_1 := \omega + \alpha_{\max}$ and $\alpha_\Sigma := \alpha_{\min} + \alpha_{\max}$. Dividing $q(s^*, m) = 0$ by s^* gives $\alpha_\Sigma m/s^* = as^* + b_0 + b_1 m$. With $ds^*/dm = -\partial_m q/\partial_s q$, where $\partial_m q = b_1 s^* - \alpha_\Sigma$ and $\partial_s q = 2as^* + b_0 + b_1 m > 0$,

$$\alpha_\Sigma \frac{d}{dm} \frac{m}{s^*} = a \frac{ds^*}{dm} + b_1 = \frac{a(\alpha_\Sigma + b_1 s^*) + b_1(b_0 + b_1 m)}{\partial_s q} > 0$$

So $p^*(m)$ is strictly increasing whenever $\sigma > \alpha_{\min} + \alpha_{\max}$.

D.7 Collapse in the published delay model

This is the proof of Theorem 5.20. Its first step is the delay version of Lemma 5.12.

Remark D.1 (The balance sheet in the published delay model). Lemma 5.12 is stated for the reduction (M), but the identity itself holds in the four-dimensional delay model (K), with the substitutions of Section 4.1, once the pupae in capped cells are counted. Let $\Pi(t) := \int_{t-\tau}^t \varphi B(s) ds$ be the number of pupae. Then $\dot{\Pi} = \varphi B(t) - \varphi B(t - \tau)$, so $\frac{d}{dt}(H + F + \Pi) = \varphi B - mF$. The term $\kappa\varphi B = \gamma_B B$ cancels the brood consumption in the food equation exactly:

$$\frac{d}{dt} [f + \kappa(H + F + \Pi)] = cF(p - p_\kappa) - \gamma_A H$$

The reading carries over. A pupa has already eaten her honey, so she counts as invested capital. The larvae are not counted, because their consumption is the flow that the pupae term books. The proof below extends this weighting to the sharper $a_\star$.

Proof. B , H and F stay non-negative, because $\dot{B} \geq -\varphi B$, because $\dot{H} = \varphi B(t - \tau) \geq 0$ on $\{H = 0\}$ and because $\dot{F} = H\alpha(f) \geq 0$ on $\{F = 0\}$. Let $\Pi(t) := \int_{t-\tau}^t \varphi B(s) ds$ be the number of pupae, as in Remark D.1. Put

$$W_D := f + a_\star(H + F + \Pi) + (a_\star - \kappa)B$$

With $\dot{\Pi} = \varphi B(t) - \varphi B(t - \tau)$ and $\dot{B} = LS - \varphi B$,

$$\dot{W}_D = F(cp - \gamma_A - a_\star m) - \gamma_A H + (a_\star - \kappa)LS + B[a_\star \varphi - \gamma_B - (a_\star - \kappa)\varphi]$$

The bracket vanishes, since $\kappa\varphi = \gamma_B$. What remains is the right-hand side of (5.18) with $a = a_\star$. The estimate of Lemma 5.14 therefore applies unchanged:

$$\dot{W}_D \leq -\gamma_A \frac{H^2}{H + v} - c(p_E - p)F \leq 0$$

All five quantities in W_D are non-negative and $a_\star - \kappa = \gamma_A v/L > 0$, so W_D bounds each of them and the solution is bounded. Integrating gives $\int_0^\infty F < \infty$ and $\int_0^\infty H^2/(H + v) < \infty$. The derivatives $\dot{F} = HR - mF$ and $\dot{H} = \varphi B(t - \tau) - HR$ are bounded, so F and $H^2/(H + v)$ are uniformly continuous and Barbalat's lemma [20, Lemma 8.2] gives $F \rightarrow 0$ and $H \rightarrow 0$. Then $\dot{B} = Ls(f)H/(H + v) - \varphi B$ is driven by a term that tends to zero, so $B \rightarrow 0$ and $\Pi \rightarrow 0$. Finally W_D is nonincreasing and bounded below, so it converges and with it f . $\square$

This is the argument of Theorem 5.17, with the brood and pupae added to the weight. Integrated from the published initial state with stores 800 g at $V = 11$, the delay model collapses within 4000 days and W_D never increases along the computed solution (numerical).

D.8 Collapse in the window while the stores stay low

This is the proof of Proposition 5.33.

Proof. Equation (5.18) with $a = a_c > \kappa$ gives $\dot{W}_c = LS(a_c - \kappa) + F(cp - \gamma_A - a_cm) - \gamma_A H$. On $\{f \leq f_{\text{crit}}\}$ the bound $s(f) \leq s^*$ holds, so

$$LS(a_c - \kappa) = Ls(f) \frac{H}{H+v} \cdot \frac{\gamma_A v}{Ls^*} \leq \gamma_A v \frac{H}{H+v},$$

and as in the proof of Lemma 5.14 the first and last terms together are at most $-\gamma_A H^2/(H+v)$. The middle term is $-cF(p_E(a_c) - p)$ and $p_E(a_c) = (\gamma_A + \kappa m + \gamma_A m v / (Ls^*)) / c = p^*$ by (5.10). This is (5.32). For the convergence, run the proof of Theorem 5.17 on $[T, \infty)$ with W_c in place of W : by (5.32) W_c is nonincreasing there, so the solution is bounded, F and $H^2/(H+v)$ are integrable on $[T, \infty)$ and Barbalat's lemma gives $H, F \rightarrow 0$. The limit $f_\infty \leq f_{\text{crit}}$ holds because $f(t) \leq f_{\text{crit}}$ for $t \geq T$. $\square$

D.9 Well-posedness, positivity and boundedness

These are the arguments behind Section 4.5.

Let

$$\Omega := \{(H, F, f) \in \mathbb{R}^3 : H \geq 0, F \geq 0, f \geq 0\}$$

Lemma D.2 (Well-posedness and positivity). *The right-hand side of (M) extends continuously to all of Ω and is locally Lipschitz there. Through every point of Ω there passes a unique solution, it exists for all $t \geq 0$ as long as it remains in Ω and the faces $\{H = 0\}$ and $\{F = 0\}$ cannot be crossed outwards.*

Proof. *Extension and regularity.* The only term not defined on all of Ω is

$$g(H, F) := \frac{HF}{H+F},$$

which enters as σg in HR and is undefined at $H = F = 0$. Since $0 \leq g \leq \min(H, F)$, it extends continuously by $g(0, 0) := 0$. Away from the origin,

$$\frac{\partial g}{\partial H} = \frac{F^2}{(H+F)^2} \in [0, 1], \quad \frac{\partial g}{\partial F} = \frac{H^2}{(H+F)^2} \in [0, 1],$$

so $\|\nabla g\| \leq \sqrt{2}$ on $\Omega \setminus \{0\}$. With continuity at the origin this makes g Lipschitz on Ω with constant $\sqrt{2}$. Every other term is smooth on Ω , so the field is locally Lipschitz. It is not globally Lipschitz, because $\partial_f(H\alpha) = H\alpha'(f)$ is unbounded in H . The remaining factors are harmless: $s(f)$ and $b^2/(b^2 + f^2)$ are smooth and lie in $[0, 1]$ for $f \geq 0$ and $H/(H+v)$ is smooth for $H \geq 0$ because $v > 0$. Picard–Lindelöf gives local existence and uniqueness.

Global existence. It suffices that the field grow at most linearly. Since $0 \leq S \leq 1$, the bounds $|LS| \leq L$ and $|\gamma_B LS/\varphi| \leq \gamma_B L/\varphi$ hold. Furthermore $|HR| \leq (\alpha_{\min} + \alpha_{\max})H + \sigma \min(H, F)$ and mF , cpF and $\gamma_A(H+F)$ are linear. Hence $\|\dot{x}\| \leq C_1 + C_2\|x\|$ on Ω and no solution blows up in finite time.

Positivity. On $\{H = 0\}$ both $S(0, f)$ and HR vanish, so $\dot{H} = 0$. On $\{F = 0\}$ the rate $R = \alpha(f) \geq \alpha_{\min} > 0$, so $\dot{F} = H\alpha(f) \geq 0$. $\square$

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